Assessing Different Heuristic Sets for Evaluating the Performance of Reading Mode Using Heuristic Evaluation – A comparative analysis

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Digital accessibility is essential to preventing feelings of exclusion in people with disabilities such as dyslexia from the physical, cultural, and digital environments that support well-being, communication and community practices. This study addresses the call for methods to evaluate the performance of reading support technologies used by people with dyslexia. The study focuses on reading mode, a reading support- and assistive technology available in most web browsers that combines features for supporting reading. Heuristic evaluation was investigated as a resource efficient alternative to traditional user testing for evaluating its performance. Two heuristic sets, Nielsens usability heuristics and Deque systems accessibility heuristics, were used for evaluating three different reading modes used on the website for Act Church of Sweden to find out to what extent they can be used to evaluate reading mode. Furthermore, the HAAT model was used to create a brief to guide the process of reviewing the problems found in the heuristic evaluations. Through a comparative analysis of the heuristic analyses outcome and capability of finding problems relating to performance of reading mode, results showed that both sets could successfully be used to find problems relating to the performance of the reading mode but that the Deque systems were better suited for diagnosing and describing found problems. Based on these results the study discusses the use of the heuristics sets and the HAAT model. The study concludes with a contribution to further research in reading support technology by proposing an adjusted heuristics set, based on the study findings, the Nielsen set and Deque systems set, for evaluating reading mode and potentially other reading support technologies.

CCS CONCEPTS • **Human-computer interaction • Digital accessibility • Assistive technology • Dyslexia • Reading support technology**

Additional Keywords and Phrases: Human-computer interaction, Digital accessibility, Web accessibility, Assistive technology, Dyslexia, Reading support technology, Heuristic evaluation

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1 INTRODUCTION

As much as 16% of people globally are experiencing a significant disability [65] that means they may face environmental and social barriers that exclude people from physical-, cultural- [6] and also digital environments such as websites. These have consequences such as a greater likelihood of unemployment, less safety and security and less community participation. The field of human-computer interaction (HCI) contributes to making digital environments more accessible to people to lessen these effects. Accessible websites require consideration of many aspects to make sure that everyone can use them, no matter what their disability might be. Dyslexia can be described as a specific learning disability that makes it difficult to write and to decode and read text [16, 19, 39]. There has been much research on dyslexia focused around supporting reading by altering the visual appearance of text or the text itself, however there is a deficit of research within HCI investigating methodologies for evaluating the performance of reading support technology for people with dyslexia [3]. In this context, performance is interpreted as the extent to which the tool supports the user in the activity of reading and could be influenced by several factors. As HCI is, and needs to be a multidisciplinary field of research [56], combining HCI practices with frameworks from adjacent research fields such as assistive technology, readability, disability studies and occupational therapy may be a way of understanding these different factors [8]. The human activity assistive technology (HAAT) model is one such framework that may help designers understand that performance of such technology can be understood by looking at the interaction into the capabilities of the *human*, the *activity* to be carried out, the *assistive technology* (AT) that allows it, and the *context* in which the activity is taking place. As such combining HCI methodologies for evaluation with

HAAT, research into dyslexia and what AT functionalities might help the individual reader [3] and help in understanding how to assess the performance of reading support technologies.

The Swedish national association for dyslexia recommends a variety of different tools that help people that have difficulties with reading [20]. Among them are reading modes, a type of reading support technology or AT available in most web browsers. They support users with reading on websites by removing distractions, simplifying layout and allowing users to tailor their visual reading experience to their specific needs [5, 26, 35, 43]. On desktop browsers, they may increase reading speed by 5%, but there is no research found that studies how they function in mobile browsers [35]. What is known about how browser reading modes function is that they inspect website source code to algorithmically determine content that is suitable for reading, to then transform those HTML elements and render it in a manner that makes it easier for users to read [35]. But despite websites having text that would be useful to read with the tool, it won't always activate. This indicates that their function is partly dependent on the coding of websites, and thus also on web accessibility.

Policymaking in the EU and Sweden in 2019 and 2025 based on the web content accessibility guidelines (WCAG), a global standard for web accessibility conformance, has or promises to make websites, products and services for the public and private sector [44] more accessible. However there are still reports of inadequate accessibility of websites in Sweden [18, 33, 51]. Industry professionals in user experience (UX) are important actors in this landscape due to their influence over factors such as visual aesthetics [38], usability [42], affective attributes [1] that contribute towards making products and services inclusive [27]. However, research seems to indicate that companies have limited time and resources to allocate towards accessibility work such as getting in touch with- and involving users in the design process [4, 27]. Automated accessibility evaluations through tools such as WAVE [62] or TAW [23] are a useful in these circumstances as it can help designers to check or monitor websites for accessibility problems according to WCAG and see what needs to be fixed. Despite the help that these tools offer, research shows that they should not be used standalone since they cannot capture and evaluate all WCAG criteria, and thus manual testing becomes equally if not more important [14, 61]. Furthermore, the literature reveals a deficit in industry professionals knowledge in accessibility and impairments and calls for improving education and training in digital accessibility [27, 48, 58]. In all, research shows that for industry work with digital accessibility, user tests, automated- and manual evaluations are important but that there is a great need for training and familiarizing professionals with accessibility practices.

Heuristic evaluations is a method that is commonly used within UX design as a resource-efficient way of finding usability problems in user interfaces without involving users, to inform the development of digital products or services. The evaluations are performed within a set scope, and the identification of problems is based on finding violations of a certain set of guidelines that is chosen before evaluations begin. Different heuristics sets can be used and choosing the right set may be integral to finding problems within the scope of an evaluation [63]. Nielsens 10 heuristics [45] are the most widely adopted set for usability, but the prerequisites for using them for accessibility problems are debated [34]. Meanwhile there are sets of heuristics that are more specialized towards accessibility, such as the Deque systems set [11]. Heuristic evaluation has broad use areas, and although it evaluators need to spend some time in preparation to become familiar with the interface and heuristics [11, 41, 63], it can be used effectively by novices and experts alike [49] to find usability problems. It does however come with a caveat that is natural when not involving users, that one cannot be sure that the problems that are found are real problems that users encounter. Approximately 20-51% of problems found in heuristic evaluations represent real problems [47]. Furthermore, while heuristic evaluations are more efficient for identifying problems, user tests usually find problems that are more severe [37]. However, it can be difficult to recruit people with disabilities for user tests [27], and heuristic evaluation may as such be an effective method to identify usability and accessibility problems in websites under and after development for monitoring purposes and with or without AT. As such the extent to which the method is useful and efficient for the purposes of evaluating reading support technology on mobile is not yet known.

The Church of Sweden believes that the Church should be open for everyone and are thus striving to make its spaces, physical as well as digital, accessible to all people. To be accessible in digital environments, the Church of Sweden needs to conform with the current digital accessibility policies in the EU [69] and Sweden [32] and are thus seeking to improve their website in these areas. As a part of striving to conform with accessibility policies, the Church of Sweden seeks to understand how people that use assistive technologies for reading view the content of their website. As the Church of Sweden website has many services and information pages, *Act Church of Sweden,* which is the Church's charity organization, and specifically their core webpages surrounding the charity, engagement and about- sections of their website is the subject for evaluation. This study would possibly give Act Church of Sweden a method for evaluating the usability and accessibility of visual reading support technologies used together with the site

to understand how users of these tools view their content. It would also supply them with documentation on accessibility and usability problems found in the evaluations carried out as part of the data collection. Because 64% of user visits and 80% of donations on the website happen on mobile devices, and little is known about the performance of reading modes on mobile, this will be the device of focus in the study.

1.1 Research aim, questions and contribution

People with dyslexia may need to use reading support technology such as reading mode to overcome barriers relating to readability and content on websites, and there is much research about helpful features and aspects of usability and accessibility that are important to support this. However, relatively little is known about how to methodologically assess the performance of such tools for people with dyslexia. As such, the aim of this study is to understand what help heuristic evaluations of reading support technology used by people with dyslexia can be in these circumstances. Because this is a topic involving HCI and accessibility, it is proposed that HCI continues the tradition of importing frameworks from other fields of research [56] such as occupational therapy and assistive technologies and thus adopt the HAAT model by Cook et al [17] to shed light on how heuristic evaluations can be performed with more sensitivity to the needs for people with disabilities. While Nielsens heuristics are the more famous and commonly used heuristics, their use for evaluating accessibility is only theorized, which means they need to be tested for this purpose. And while the Deque heuristics are specialized towards accessibility, they appear not to have been tested in research and neither for the purposes of evaluating assistive technologies such as reading mode.

To find out how to evaluate reading support tools, the following primary research questions has been formulated:

RQ1: To which extent can Nielsens 10- and Deque systems heuristics be used to evaluate browser reading support technology such as reading mode?

Additionally, one secondary research question is formulated:

RQ2: What can evaluator descriptions of the four components of the HAAT-model add to the reviewing of problems found in heuristic evaluation of reading mode as used by people with dyslexia?

This study will be a contribution towards filling the gap in research regarding what methodology may be used to assess the performance of these types of tools specifically for people with dyslexia [3]. The contribution is believed to be threefold: (1) it will provide a methodological discussion as to the means to evaluate the performance of reading mode through heuristic evaluations with usability and accessibility heuristics, (2) it will describe how the method can be used as a means for researchers and design professionals to become more knowledgeable in the assessment of web accessibility and the use of assistive technology and (3) it will provide a set of heuristics to use for evaluating reading mode and similar reading support technology.

1.2 Delimitations

There are many aspects of heuristic evaluation that may be studied, this study will focus primarily on what heuristic sets to use, what problems they can find, as well as how the evaluation itself should be carried out based on the user's need and the target of evaluation. It is put into context by Alonzo and Hassan [3] call for methodologies that can evaluate the performance of reading support tools for people with dyslexia. Additionally, the study is limited to using reading mode on a selection of pages from the Act Church of Sweden website, which means that the results will be limited to the context of this website, and is aware that the tool might perform differently on other websites.

2 RELATED WORK

2.1 Reading difficulties

People with sensory and cognitive impairments such as BLV, dyslexia and people with low literacy may be put in difficult positions when met with barriers associated with the readability of user interfaces. Barriers may exclude people from social and cultural environments [6] and can make social connections and participation in communities difficult [17]. People with dyslexia experience difficulties with reading and writing and these difficulties may vary in severity from person to person. In many cases it entails problems with decoding words, spelling, word recognition and recall, and impacts reading speed and accuracy as well as self-perception and self-esteem [3, 19, 31, 39, 55]. Readability encapsules many different properties of text such as language use, typography and the medium through which to make it available to readers [8]. Similar to HCI's focus on the user and assistive

technologies focus on the human [17], readability concerns itself with who the readers are and why they read [8]. It can be discerned that reading has dimensions, the *temporal* axis that concerns reading over different windows of time and the *purpose* axis that concerns reading with different motivations in mind [8]. In regard to the present study into reading mode use on Act Church of Sweden, the interpretation is that a reader will read for the purpose of *content uptake* to learn about processes such as how to donate or get involved in the community, and in regard to time that they will do *glanceable reading* and deep reading to find and learn for example about how to become engaged in the community or to find out if Act Church of Sweden is the right organization to donate to. Along with HCI and assistive technology, much research of readability has gone into what can be done to adapt user interfaces to support different kinds of reading thus alleviate some of the difficulties people with dyslexia experience, such as: changing font, word- and character spacing, size, background color, line width and spacing and text alignment [3, 8, 54, 55]. These suggested interface adaptations are primarily concerned with how text is visually presented, but some research advocate for changes to the content itself to be another solution, such as using visualizations and plain language that is easy to understand [3, 57]. For people with dyslexia, it is also common to use assistive technologies such as screen readers, text-to-speech software, alternative keyboards or screen magnifiers to support reading, whether it be for reading physical or digital books, or websites.

2.2 Reading support technologies

Through a 25 year analysis of HCI research on reading support technologies, Alonzo and Hassan [3] states that most research on the topic concerns BLV (blind and low vision) users and users with dyslexia, and even though digital reading has been thoroughly explored within HCI, there has been a lack of research on reading support technologies over the last 2 decades. The problems that limited accessibility of digital products and services pose to people with reading and writing difficulties is covered and suggestions on how to alleviate these vary from technical innovations to UI adjustments [3]. Screen readers like JAWS or NVDA are assistive technologies that is widely used for example by people with BLV or dyslexia [3, 39] to have images and textual digital information read aloud. In this study, the reading support technology to investigate is reading mode, a tool used for reading that is built into web browsers like Safari, Chrome, Firefox and Microsoft Edge. These tools typically offer a broad variety of features for making website text easier to decode by providing features that are known to support reading, such as: allowing focus through reducing distractions, editing typography, background color, layout preferences and read aloud with our without text highlighting [3, 8, 26, 40, 54, 55].

Reading mode would not be described as assistive technology in a traditional sense as it is part of a web browser, it is what could be described as a mainstream user agent, and is not built to support someone with a specific impairment but to support a diverse audience [17, 66]. This distinction, although not absolute [66], entails that reading mode may be exempted from the WCAG. There seems to be no official resources from the world wide web consortium (W3C) that specifically target these tools and the World health organization (WHO) does not mention specifically reading mode as a priority product, but both screen readers and screen magnifiers whose functionality often are included in reading mode, are [64]. Reading mode could instead be described as an adaptive strategy [28] or an assistive tool, and is for example listed as one of the recommendations of tools that supports focused reading by the Swedish dyslexia association [20]. As such, in this study reading mode will be studied as if it is an assistive technology on the grounds that it is used by a person with dyslexia. When referring to reading mode in this study, the phrases reading support technology and assistive technology (AT) will be used.

The distinction between mainstream user agents and assistive technology does not mean that reading modes aren't affected by coding practices. They are, as assistive technology, in many cases dependent on information being programmatically determined, meaning they must be correctly coded so that information can be extracted and later presented to users through these tools [66]. Images must for example be marked with appropriately descriptive <alt> tags, elements must be labeled correctly and headings must be hierarchically correct to help users make sense of information [66]. Reading mode has, however, not been studied enough to understand how they render website content or how they support users. An exception is a study by Li et al [35] that looked into how the use of Mozilla Firefox Reader View on a computer desktop impacted reading speed and user experience. In looking at how the Reader View changes websites they could discern that it algorithmically determines web content that are suited for reading and removes for example advertisements, menus and logos, but keeps and renders text blocks and images for reading [35]. Additionally, they found that Reader View was only available in 2% of homepages and 41% of child pages out of the 100 they sampled. Despite this, they found that Reader View significantly improved the reading speed, perceived readability and aesthetics for people with and without dyslexia [35]. What could be further studied is how these- and facets of web accessibility affect reading mode performance on mobile devices, that in comparison to their desktop counterparts have less features. While many studies cover the

benefits of these features, few investigate how they perform on mobile devices [3], and no found literature addresses what factors determine how well they will function on mobile devices or how to evaluate this. But in finding that the reading mode functionality is dependent on the coding of websites, looking into web accessibility practices is necessary.

2.3 Evaluating web accessibility

The term web accessibility refers to how websites, tools and technologies are designed so that people with disabilities can use them [29]. The web content accessibility guidelines (WCAG) exist so that the content of products and services can be made accessible to a wide range of users with disabilities, such as blindness and low vision, limited mobility, learning disabilities and cognitive limitations [66]. The WCAG is based on four main principles called POUR which all relate to how content and user interface (UI) components such as text, search fields, links and images are presented and made accessible. *Perceivable*, regarding the presentation of content and UI components such as text, search fields, links and images to be perceivable, *Operable*, regarding how UI components and navigation must be usable, *Understandable*, regarding how information and the use of the UI must be able to be understood, and *Robust*, regarding how content must be able to be interpretable and compatible to many user agents as well as assistive technologies [66], such as reading mode. There are many reasons for companies and organizations to design for conformance with WCAG, perhaps foremost is that the guidelines have become part of policies in many countries through the EN 301 549 (2019-11) standard. In Sweden, it is mandatory for web pages for the public sector to follow this standard through Act (2018:1937) on the Accessibility of Digital Public Services, and for the private sector through Act (2023:254) on the availability of certain products and services. Not complying with the basic requirements for WCAG conformance would not only mean reaching less users but can also result in financial repercussions for companies not conforming to policies. Secondly, they provide a ready framework for designers to design for accessible experiences that benefit everyone [57].

A formal evaluation of WCAG conformance may be carried out within a certain scope such as a few central web pages using the web content accessibility guidelines – evaluation method (WCAG-EM) to generate a report containing observations of violations with each of the accessibility guidelines listed under each of the four main principles. WCAG-EM is often specific to evaluating websites in a comprehensive and detailed manner that is not suitable for quick evaluations. Because of this, the web accessibility initiative (WAI) recommends evaluators to do a quick checklist evaluation to affirm if a website is in need of a full evaluation [30]. But despite quick or comprehensive WCAG evaluations showing that all criteria are met, the content might still not be accessible to users with disabilities [29, 50, 59]. This is a known issue in research mapping the adoption of accessibility practices in the industry, which shows that the main drive for accessibility adoption for UX professionals are laws and policies [27] despite research showing that there is also an economic, societal and quality of life benefits to it [13, 17, 29]. This affirms the need for knowledge of web accessibility and impairments among designers, and that it may need to be complemented with good usability and UX to become accessible.

Where formal evaluations are resource demanding, automated systems that monitor accessibility are another way for IT companies to keep updated on the accessibility of their websites while freeing up resources to work on correcting errors [14, 58]. A downside of these systems that may make them unlikely to replace manual evaluations is that their developers may struggle to keep up to date with WCAG [14, 61], as well as being insufficient as a standalone inspection of WCAG compliance [58]. This points to the need for manual evaluations that are easier to perform.

2.4 Heuristic evaluation

A heuristic evaluation can be carried out as part of the iterative design process where it serves as a fast and cheap method to provide formative evaluations of prototypes in different stages or summative evaluations of interfaces [63]. The evaluation is carried out by a usability or subject expert following a certain set of tasks while looking for usability problems. More commonly a team of three to five experts from different domains and experience levels are included, which contribute to finding problems that are more likely yield reliable results [46] and real problems [2, 63]. The evaluator notes down found problems, what heuristics they refer to and what severity level it has from a scale of 0 to 4, where 0 marks no problem and 4 a catastrophic failure [46]. There are many different sets of heuristics, and Nielsen's [45] 10 heuristics are often referred to as the most comprehensive list to use [25, 63]. Heuristics are shown to be developed by several different methods, like literature reviews, user studies, expert interviews or mix of them [52]. It is important to note that heuristics are a representation of greater knowledge that is attained over time, through studies and practical experience, meaning that even though the evaluator may for example be a usability expert they may be lacking knowledge in

accessibility practices that heuristics cannot wholly replace. Furthermore, heuristic evaluations cannot wholly replace the capability of WCAG-EM in finding accessibility problems and guideline violations but might be a way to contribute towards websites are accessible as well as having a high usability and a good user experience.

2.4.1 Nielsens 10 heuristics

Research reveals different arguments over Nielsen and Molich's set of heuristics and its role in evaluating accessibility. Studies establish that there is an overlap between WCAG and usability heuristics [15, 42] that allows usability heuristics like Nielsen's to be used to evaluate accessibility of websites (see table 1). However, Khasawneh et al [14] argues that for heuristics to be usable when evaluating accessibility, they have to expand from Nielsen set to cover such things as captions and audio descriptions, keyboard operability, readability for screen readers and improved error recovery. Furthermore, heuristics used in this manner should be derived from accessibility standards and guidelines such as WCAG or plain language [32, 34, 66]. This seems to mean that although it is doable, it is not enough to interpret Nielsens heuristics differently, instead there needs to be new sets of heuristics to appropriately inspect accessibility as per the heuristic evaluation method.

Table 1: Overlap between Nielsens 10 heuristics and WCAG according to Moreno et al [42] and Casare et al [15]

Heuristic #	Nielsen's 10	WCAG Percievable	WCAG Operable	WCAG Understandable	WCAG Robust
Heuristic 1	Visibility of System Status	X	X		X
Heuristic 2	Match Between the System and the Real World	X	X	X	X
Heuristic 3	User Control and Freedom	X	X	X	X
Heuristic 4	Consistency and Standards		X	X	
Heuristic 5	Error Prevention			X	X
Heuristic 6	Recognition Rather than Recall		X	X	X
Heuristic 7	Flexibility and Efficiency of Use		X		
Heuristic 8	Aesthetic and Minimalist Design	X	X		
Heuristic 9	Help Users Recognize, Diagnose, and Recover from Errors	X	X	X	X
Heuristic 10	Help and Documentation		X	X	X

2.4.2 Deque systems accessibility heuristics

Despite differing views of the use of Nielsens heuristics for evaluating accessibility, there exists few sets of heuristics specialized towards web accessibility. Boudreau and Pasi of Deque systems have created a set of accessibility heuristics to be used by designers. The heuristics represent the technical requirements found in WCAG [10, 11] and as such provide designers with a resource for designing and evaluating for web accessibility without needing to read through the WCAG. Their ties to WCAG are not described in literature but appear clear, and because they are built upon Nielsens heuristics they include both guidelines for general usability such as error prevention or consistency as well as to guidelines in POUR, such as Contrast and Legibility mapping to *Perceivable*

and *Robust*, or Navigation, Interaction methods and Timing and Preservation mapping to *Operable* [11, 66]. They thus seem to be a better fit for evaluating reading mode but appear yet untested in research.

2.5 Research gap

Related work shows that while much is known about what features can support people with dyslexia in reading, little is known about to what extent they provide support for users with dyslexia. This is defined in literature by the performance of the reading support technology [3], which also can be referred to as assistive technology. Reading mode is one of these technologies, and one that has been studied little despite it being recommended for use by the Swedish dyslexia association [20] and it being available in the majority of web browsers. Furthermore, reading mode has a tie to web accessibility in how its function is dependent of the coding of websites, meaning that web accessibility guidelines such as WCAG are required to be able to diagnose the performance of the tool. But since there is no guarantee that WCAG conformance entails accessibility through these tools, it is necessary to also involve usability. While user studies are effective in finding and diagnosing usability issues, it can be difficult and time consuming to find representative users and to plan user tests both within research and industry. As such, to be able to make a research contribution to HCI regarding the methodological process of evaluating the performance for such tools, the use of heuristic evaluation is proposed as a quick and less resource requiring alternative to user studies. But as the effectiveness of the method is dependent on the heuristic sets used, both heuristics that may be suited for this task and the way of applying them for these purposes must be studied.

3 THEORY

This study investigates the usefulness of Heuristic evaluations for measuring the performance reading mode through the lens of the human activity assistive technology (HAAT) model. The model stems from- and is most widely used and valued in the field of occupational therapy [24] and for occupational therapists to help people choose what AT is best suited for them given their disability and the activities they need to perform in daily life [17]. However it may also be used for product research and development as well as product usability studies [17] for example of intuitive interfaces [36] where learnability, efficiency, memorability, errors, satisfaction and ease of use as key features to study. The HAAT model is used to rule the extent to which assistive technology can help people with communication, cognition, mobility and/or manipulation by considering the human, the activity, the assistive technology as three equally central components fit within a context [17, 24]. The *human* component encompasses the users' abilities, for example sensory or cognitive, the user's role in life using technology in activities and motivation to use AT. The *activity* component is defined by the execution of tasks that users come across in life, it might for example be related to productivity, leisure or necessity in addition to how activities play out over or at the same time, in space and place. The *context* component includes physical context like surroundings, social context like with peers and strangers, cultural context such as communities and religious practice and institutional context such as legislations, regulations and policies affecting funding for device acquisition and rights [17]. Finally, the *assistive technology* component is viewed as what enables a human to do an activity in a context, for example that wheelchairs allow movement for the physically impaired or screen readers allowing people with BLV to read.

Applications of the model in research outlines its use for providing a background context, informing study design or incorporating its conceptual elements for interpretation of findings [24]. Within HCI, the model has been used to validate questionnaire design [67] and interview questions for domain experts [60] in AT studies by covering all four components, thus making sure that most of the important factors for performance is covered. Similarly, this study seeks to cover all four components, which can be achieved by writing a brief that describes the four components based primarily on literature on dyslexia and readability, evaluation results and the context of the Act Church of Sweden website. This brief will guide the review of found problems in absence of domain experts, users and additional evaluators. When it comes to heuristic evaluations, while there is a framework such as WCAG that explains what usability, accessibility and subsequent web content compatibility mean for AT, it does not explicitly focus on factors influencing the interaction like the HAAT model does. As such, the model could complement the methodological process and heuristics of heuristic evaluation. This would help evaluators reiterate the importance of the user with a disability, and to break down the interaction into activities (what the user wants to do and achieve), AT (how technology helps them achieve it) and context (the website) to prompt consideration of how such technologies mediate interaction on webpages and thus inform the problem finding and severity rating of the problems that may be encountered in evaluations. The model sees performance as the

outcome of how well humans can carry out specific activities with the AT. However, this study means to first assess the performance of the reading modes. With reading mode, activities performed with the tool are meant to foremost support reading and comprehension. They accomplish this by offering users alternatives to how information is presented. Looking at reading mode through the lens of HAAT can thus provide a way to understand how well the tool supports reading and comprehension in the context of its use. This complements the way heuristic evaluation looks for violations of usability guidelines by considering how the user with dyslexia uses reading mode on websites to read and understand information.

In regard to this study, each of the components of HAAT are described as follows: the human with dyslexia, the activity of reading and comprehension, the reading mode as the AT and the website and device it is used on as the context. As in the real world, this context comes with many of its own norms and knowledge, for example how a floppy disk represents the save function. All of these components make up an understanding of how AT may support people and may be beneficial to consider in evaluations concerning digital accessibility. In regard to Act church of Sweden, the activity of reading the website content with AT be defined as learning and application of knowledge towards participation in community and civic life aspects of activities mentioned by Cook et al [17], seeing as for example reading about becoming an engaged member in Act may result in becoming part of the community and meeting other people that are engaged in Acts' cause.

4 METHODOLOGY

This section describes the methodology of the study and goes over the method for data collection, data analysis, positionality and ethics. The general outline of the study is described in figure 1.

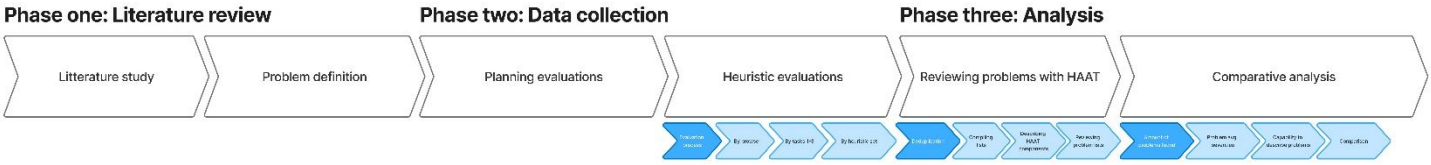


Figure 1: overview of study phases

4.1 Heuristic evaluation as a data collection method

Since this study investigates heuristic evaluation as method of evaluating the performance reading modes, it is also the method of data collection. As no heuristics specialized towards reading assistive tools have been found, the heuristics used in the evaluations was Nielsens 10 heuristics [45] and Boudreau and Pasi of Deque systems [11]. Additionally, since the target assistive technology is made to support users with visual reading, the heuristic evaluation is applied to a graphical user interface (GUI), unlike assistive technologies like screen readers which rely mainly on keyboard and audio. Despite the differing views of the interpretation and use of Nielsens [45] and Molichs [47] heuristics for accessibility evaluations [15, 34], this set was used because of its relatively unexplored use for evaluation of user interfaces of assistive technologies such as web browser reading mode. The case is similar with the Deque systems set which inherently is tied to WCAG [11] but appear unexplored for evaluating assistive technologies.

These two heuristic sets were used when conducting heuristic evaluation of the assistive tools. Because heuristic approaches found in literature come with their prescribed rules, these rules have been kept in the evaluations of this study. Additionally, to evaluate the tool performance for reading important information, the evaluations were set up with tasks representing the core functions and pieces of information on the Act Church of Sweden website. The severity rating used was inspired from Nielsen [46] who originally proposed a scale from 0-4, where 0 is a non-problem and 4 is a catastrophic failure where users can't move forward in the interaction. However, in the evaluations a scale from 1-4 was used, omitting the non-problems until problems are reviewed. If the evaluation was conducted by several evaluators, it would be suitable to reevaluate the severity ratings as a group and in that case assign a 0 rating. The tasks that were followed for the heuristic evaluation were to read about: donating to charity, about the cause of Act Church of Sweden and how to become an engaged member in the community, the tasks are expanded upon in the appendix (see appendix A).

Each task was carried out systematically in a fixed indoor space using three different in-browser visual reading support tools on an iPhone 14, Safari, Chrome and Firefox, representing three of the top 5 most used browsers on the website the latest year. For ease of documentation, a template (see Appendix B) was used as a guide to collect qualitative data of problems found and included an ID, short description, problem location, what heuristic was violated, a severity grading, problem reasoning, and screenshots.

4.2 Alternative methods for this study

It is recognized especially in accessibility and AT studies [17] that involving users in testing is a responsible and effective way to develop tools and interfaces that are user centered. However, it can be difficult to recruit users with disabilities for user tests, and the tests demand careful planning and resources [27]. An alternative way of involving users in a manner that requires less resources are to adopt participatory approaches and include them in evaluations. In such methods, referred to as Barrier Walkthroughs, users are effectively considered as an evaluator and might thus help in finding real problems as opposed to theorized ones and thereby increasing the reliability of results [7, 12].

4.3 Data analysis

To answer the research question of how which out of the Nielsen, and Deque systems heuristics sets are suited for evaluating the performance of reading mode, a qualitative comparative analysis was conducted based on two aspects of the gathered data, (1) the conceptual differences and similarities between the two sets of heuristics used and their capacity to help evaluators find problems in the context of the study, and (2) the categorization of problems found in the heuristic evaluations of three different reading modes used on the Act Church of Sweden website, based on severity and heuristics documented. Comparative analysis is used to understand differences and/or similarities between different objects of interest in research, such as settings and interviews [68] or as in this study, between heuristic sets. This enables researchers to understand familiar practices from different perspectives, allow testing of theories or discover alternative methods [21]. It requires objects of comparison to be clearly described and sufficiently comparable in terms of for example concepts and how they are measured [21, 22]. Comparisons of different heuristics sets or methods is present in related works within HCI, for example comparing Nielsen's set and WCAG [15, 42], heuristic sets versus user studies ability to find usability problems [37] or evaluator profiles in heuristic evaluation [49]. The approaches to comparison are quantitative, qualitative or mixed. Paz et al [49] uses an approach for quantitative heuristic evaluations with a 60-item checklist with yes, no, neither or N/A answers that are quantified to be able to assess a final scoring between 0 and 60 on product usability, allowing for comparison of scores between evaluators of different background. Casare et al [15] instead seemed to adopt a qualitative approach to comparison between Nielsen's set and WCAG by analyzing how they are defined. This study adopts a similar qualitative approach to comparing heuristics, looking not only at definitions but also at the concepts of usability and accessibility that they cover. Quantitative data is used descriptively to support comparisons of found problems and their severities.

The heuristic analysis of each reading mode using the two heuristic sets was processed as described in literature [63]; problems found were deduplicated and compiled into a single list for each of the reading modes, severity ratings were reviewed and then organized problems in a way that is useful, namely by severity. Importantly, the problems were reviewed only from the perspective and experience of the author, using HAAT as a theoretical lens. As such, data could not be corroborated with findings of other evaluators, as is usually done in heuristic evaluation [41, 63].

After deduplicating and compiling lists of problems found in the evaluations, the HAAT model was used as a theoretical lens to review the severity of problems by each heuristic set. The HAAT model does not provide the information to interpret results on its own, such as what AT is suited for reading on a website, but serves as a framework by which to do so, and the use of it informs what requirements should be imposed on such technologies so that they can best support the user. As such, to begin this process each of the four components of HAAT were described (see Appendix J): a person with dyslexia (human) using reading mode (AT) to support the activity of reading (activity) information on the website for Act Church of Sweden (context). Contrary to how previous HCI literature has informed its use on questionnaire or expert interviews the model to guide data collection [60, 67], these descriptions were importantly informed by literature on readability and dyslexia, the website of Act Church of Sweden, the three tasks (see Appendix A), the results from the heuristic evaluation and literature on reading mode and thus does not reflect an actual use context. In relation to the second research question, this use of the HAAT-model was compared with the nature of reading mode as a mainstream- or assistive technology to understand how the model fits into heuristic analysis and discuss how that may help researchers and professionals take the needs of people with disabilities into consideration in research and industry settings.

After reviewing and organizing the problem lists from the two heuristic analyses by severity (see Appendix E) and problem categories (see Appendix F) using each heuristic set, descriptive statistics were generated which allowed for a comparative analysis of the following criteria: the number of problems found, their average severities, evaluator interpretations of heuristics and conceptual overlap (see Appendix K), which heuristics used to describe what problems, and the categories of problems that could be found.

4.4 Positionality statement

This study is motivated by the Church of Sweden wanting to know about the use of assistive technologies such as screen readers, ReadSpeaker [53] and reading mode on their website, to be able to improve the accessibility of their websites. The lists of problems found in the heuristic evaluations represents this motivation, and the use of the method of heuristic evaluation for the purpose of evaluating reading mode is a possible way for the Church of Sweden to monitor accessibility related to similar tools in the future. The author was the sole evaluator in the study and do not have lived experience of dyslexia, but have studied the subject of dyslexia in regard to digital accessibility, WCAG and heuristics in a bachelor's degree project [9] and thus has experience in these domains that can influence the ability to find problems in the heuristic evaluations [47].

4.5 Ethics

It has been established that domain knowledge and experience is a factor that influences the result of heuristic evaluations [12, 63] and development of assistive technologies [2, 17]. Since the author is the sole evaluator in this study, the evaluator's previous experience with the subject become crucial to the trustworthiness of the results [22], it is disclosed here. The evaluator has no personal experience of having reading difficulties like dyslexia, but have done studies on the subject of dyslexia, digital accessibility and heuristics before [9]. Furthermore, as heuristic evaluation is typically done by a usability or domain expert and not a user, the tasks of the inspection included when to use the assistive tool (see Appendix A). However, it is acknowledged that the use of the tools described in the task may not be an accurate depiction of how a real user would interact with the tool. Additionally, accessibility is an important matter because the well-being and inclusion of many people are on the line. Considering this, making assumptions about the usability and accessibility of these tools from a theoretical point of view can be regarded as unethical with the amount of domain knowledge the author possesses, despite earlier experiences. The involvement of literature and HAAT in this study is an effort to compensate for these assumptions.

5 RESULTS AND ANALYSIS

For ease of explaining heuristics in this chapter, the following abbreviations will be used when the heuristics are mentioned: N10H1 for example stands for Heuristic 1: Visibility and system status in Nielsens' set, and DQH1 stands for Heuristic 1: Interaction methods and modalities in Deque systems set. Below is a table that gives an overview of the heuristic sets (see table 2).

Table 2: Heuristics sets

Nielsens 10 (N10)	Deque systems (DQ)
H1 Visibility of System Status. Designs should keep users informed about what is going on, through appropriate, timely feedback.	H1 Interaction methods and modalities. Users can efficiently interact with the system using the input method of their choosing (i.e. mouse, keyboard, touch, etc.).
H2 Match Between the System and the Real World. The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon.	H2 Navigation and wayfinding. Users can easily navigate, find content, and determine where they are at all times within the system.
H3 User Control and Freedom. Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted state.	H3 Structure and semantics. Users can make sense of the structure of the content on each page and understand how to operate within the system
H4 Consistency and Standards. Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.	H5 Error prevention and states. Interactive controls have persistent, meaningful instructions to help prevent mistakes, and provide users with clear error states which indicate what the problems are and how to fix them whenever errors are returned.

H5 Error Prevention. Good error messages are important, but the best designs prevent problems from occurring in the first place.	H5 Contrast and legibility. Text and other meaningful information can be easily distinguished and read by users of the system.
H6 Recognition Rather than Recall. Minimize the user's memory load by making elements, actions, and options visible. Avoid making users remember information.	H6 Language and readability. Content on the page can easily be read and understood by users of the system.
H7 Flexibility and Efficiency of Use. Shortcuts, hidden from novice users, may speed up the interaction for the expert user.	H7 Predictability and consistency. The purpose of each element is predictable, and how each element relates to the system as a whole is clear and meaningful, to avoid confusion for the users.
H8 Aesthetic and Minimalist Design. Interfaces should not contain information that is irrelevant. Every extra unit of information competes with the relevant units of information.	H8 Timing and preservation. Users are given enough time to complete their tasks and do not lose information if their time (i.e. a session) runs out.
H9 Help Users Recognize, Diagnose, and Recover from Errors. Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.	H9 Movement and flashing. Elements on the page that move, flash, or animate in other ways can be stopped, and do not distract or harm the users.
H10 Help and Documentation. It's best if the design doesn't need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.	H10 Visual and auditory alternatives. Purely visual or auditory content that conveys information has text based alternatives for users who can't see or hear.

5.1 Evaluation overview

Each problem found in the evaluations was documented with an ID, short problem description, location on the website and/or in reading mode, corresponding heuristics, a severity grading and a longer problem description. Table 3 and 4 present overviews of selected data gathered through the evaluations, represented with descriptive statistics of data from the evaluations (see whole overview in Appendix C). It shows an overview of each heuristic set, how many total- and post reviewing unique problems that were found using them, their average severity scorings, as well as an overview of each Reading mode evaluated using them.

After reviewing problems with HAAT and declaring non-issues and removing duplicates, a total of 34 unique issues were found using Nielsens 10 heuristics and a total of 38 unique issues were identified with the Deque systems accessibility heuristics. Most of the overall problems as well as unique problems were found when evaluating the reading mode for Mozilla Firefox, second to Safari and then Google Chrome. The average severity of the unique problems was calculated based on the severity scoring of every reported problem of similar description going into the unique problem. The problem "*Reading mode does not render all content on the webpage, important information is lost*" that was evaluated with Deque systems in Firefox reading mode was for example supported by 11 problems across all tasks and had an average severity of 2, marking it as a minor problem according to Nielsen's severity grading that both heuristic sets share. The corresponding unique problem also found in Firefox reading mode *"Reading mode does not render all content on the web page"* as evaluated with Nielsens set was supported by 8 problems across all tasks, also with an average severity scoring of 2, a complete overview over these categorizations can be found in Appendix F.

Table 3: Descriptive statistics of heuristic evaluations

	N10	DQ
Total problems found	116	90
Total unique problems	35 (34)	41 (38)
Unique problems declared non-issues	1	3
Overall most mentioned Heuristic of total problems	H4 (77)	H7 (61)
Overall least mentioned Heuristic of total problems	H10 (3)	H10 (1)
Main problem categories		
Content rendering		
Total problems found	34	42
Unique problems found	6	12
Mean severity	1,7	2,2
Limited navigation		
Total problems found	15	10
Unique problems found	6	7
Mean severity	2,1	2,9
Tool availability		
Total problems found	12	13
Unique problems found	6	9
Mean severity	1,8	2,0
Missing or poorly working features		
Total problems found	18	12
Unique problems found	11	6
Mean severity	2,8	2,6
Safari total problems found	37	28
Unique problems found	18	18
Mean severity	2,1	2,2
Task 1	16	11
Task 2	8	11
Task 3	13	6
Chrome total problems found	37	25
Unique problems found	15	12
Mean severity	2.0	2,1
Task 1	21	8
Task 2	7	6
Task 3	9	11
Firefox total problems found	42	37
Unique problems found	19	18
Mean severity	1,8	2,6
Task 1	17	14
Task 2	11	14
Task 3	14	9

*Rounded to 1 decimal position to reflect severity gradings used

Table 4: Number of unique problems and their severity averages

	Safari N10	Safari DQ	Chrome N10	Chrome DQ	Firefox N10	Firefox DQ	Total unique N10	Total unique DQ
Total problems found	18	15	15	12	19	18	34	40
Cosmetic problem	3	3	4	1	5		6	4
Minor problem	11	9	8	9	10	9	19	22
Major problem	3	6	2	2	3	8	7	13
Catastrophic	1		1		1	1	3	1

5.2 Comparing differences and similarities in heuristic sets

It has been reported that Nielsens heuristics can be mapped to WCAG [15], the amount- and types of problems found by each heuristic sets pertaining to the accessibility supported by the reading modes show that Nielsens heuristics perform similarly to Deque Systems heuristics when finding problems related to performance of the reading modes. The major differences found in the comparative analysis of the heuristic descriptions show that the connection with accessibility is more obvious with the Deque systems set than with Nielsens set, but when it comes to assistive tools like reading mode both heuristic sets are lacking in a few regards. Nielsens heuristics are based on usability engineering studies and thus envelops many fundamental aspects of usability in user interfaces: users must be able to recover from errors, errors should be prevented, the language used should be easy to understand, interfaces should not contain any unnecessary elements, and so forth. The Deque systems heuristic set however is synthesized from technical requirements in WCAG and presented in such a way as to be usable by designers [11] and is therefore specialized towards accessibility. These two sets cover usability and accessibility of interfaces independently, but the comparative analysis and use of them in the evaluations found that their capacity for describing problems overlaps conceptually when evaluating reading mode. Below are results of the comparative analysis of heuristics found to be useful in heuristic evaluations of reading mode. The following subsections will explain differences and similarities of heuristics that were important to the explanation of problems in the evaluations, and how they may help to assess the performance of reading mode going forward, they are: *Consistency, use of language and readability, encountering errors, navigation, content structure and hierarchy as well as efficiency*.

Table 5: Overview of problem categories with associated heuristics and average severities

Heuristic sets, severity/categories	Information loss upon rendering of content in reading mode	Limitations of navigation	Tool availability	Helpful features missing or working poorly
N10	34 problems (6 unique)	15 problems (6 unique)	12 problems (6 unique)	18 problems (11 unique)
Avg. severity*	1.7	2.1	1.8	2.8
H1	4	3	5	11
H2	2	0	0	2
H3	0	5	4	3
H4	33	12	5	6
H5	2	6	4	8
H6	3	2	0	3
H7	4	4	7	5
H8	6	0	0	0
H9	0	3	1	9
H10	0	1	1	1
DQ	42 problems (12 unique)	10 problems (7 unique)	13 problems (9 unique)	12 problems (8 unique)
Avg. severity*	2.2	2.9	2.0	2.6
H1	2	1	7	7
H2	20	4	7	2
H3	22	0	2	2
H4	0	4	2	2
H5	3	0	0	4
H6	12	0	1	8
H7	38	4	6	4
H8	0	0	2	2
H9	1	0	1	1
H10	0	0	0	1

*Rounded to 1 decimal place

5.2.1 Consistency

Heuristics N10H4 and DQH7, both relating to consistency and what users can expect from an interface, were used extensively for describing problems across the different evaluations (see table 5), with most violations related to rendering of content in reading mode. N10H4 was represented in all but one of the 33 problems listed in this category, and DQH7 was represented in 38 out of 42 of these problems. They are both used on the basis that when content is rendered in reading mode, it does not include sufficient content that would make the rendered content consistent with the website content, and as such the user wouldn't be able to see or read that content on the page. And if the content isn't rendered, users that have difficulties reading on the site would be forced to read in an environment not suited for them. This speaks to heuristics regarding consistency of function and user interfaces being useful for evaluating reading mode. Another interpretation of these heuristics were interface elements being placed consistency around the interface. This was observed as an issue with the menu settings in Safari that moved around based after using a function, the button for adjusting text size would for example pop out from the menu settings to a new location in the browser window after

interacting with the decrease/increase in the menu settings. If it were to remain in the menu settings it would have allowed consecutive inputs in and users would not have to go into the menu again to access the same setting. This discrepancy in the use of consistency heuristics shows that the interpretation of these heuristics can be broad and its meaning might have to be made precise for it to be accurately used in evaluations.

5.2.2 Use of language and text readability

Next, N10H2 Match between the system and the real world is conceptually akin to DQH6 Language and readability, both being about using a language that the user can understand. DQH6 which also mentions readability and that content on a page should be easily read, has a strong conceptual connection with literature on how text adjustments may aid people with dyslexia [3, 54, 55], and thus are theoretically better fitting to use for evaluating reading mode, which ideally should allow users to make such adjustments. DQH5 Contrast and legibility are similarly to DQH6 divided into two aspects, with legibility being tied to readability in WCAG criterion 1.4.8 Visual representation and should be able to be adjusted in browsers or other user agents such as reading mode. Contrast however may be tied to contrast ratios in WCAG and are, although contrast between background and foreground (text) is also useful as an interface adjustment, conceptually related to disabilities such as blind and low vision (BLV). As such these three heuristics related to text and its presentation may be of use in evaluating reading mode, but it may be necessary to divide them into one that is about the use of language, as N10H2, and one about text readability. However relatively few of these heuristics were used in the evaluations, owing to that there were very few problems encountered having to do with language used, except for where it wasn't present for describing- or giving user feedback on features that are difficult to understand from just a symbol (see figure 2). From a HAAT perspective, looking at this audio modal and knowing it's related to content being read aloud one might ask what exactly is being wound back? And how much? For a user with dyslexia performing the activity of reading a web site with this tool, it is important that they can understand and predict how it works, especially since there are no highlighting of text that can give users feedback on the action as the audio is being played. Users may be accustomed to these symbols being used in other audio or video players to go back/forwards 10 or 15 seconds when it actually goes back/forwards one sentence at a time. This shows a mismatch between contexts where one is culturally related to music or video, and the other functionally relevant to these specific assistive technologies. As such, evaluations looking out for how features are explained through visuals and text may be useful in interpreting usability problems in reading mode or similar assistive tools.

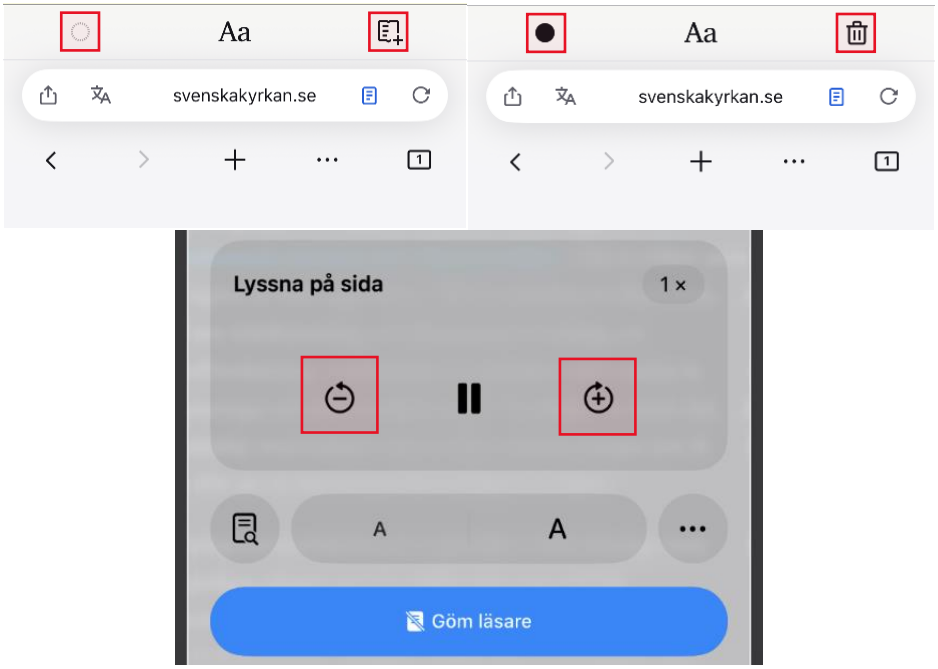


Figure 2: symbols for reading list in firefox (top), and read-aloud forwards/backwards in safari (bottom) features that may be hard to understand visually

5.2.3 Encountering errors

The comparative analysis found N10H5 and N10H9 to be conceptually similar to DQH5 as both deal with users encountering errors. Although few errors were encountered in the evaluations, they are a ubiquitous part of assessing the usability of any user interface. That said, DQH5 encapsulates both meanings of N10H5 and N10H9 as well as covering states of errors, which makes it more useful than the N10 heuristics.

5.2.4 Navigation

Nielsens heuristics have no specific guidelines that cover navigation. This made it difficult to assess problems with navigation despite being aware of their existence. Focusing on the human with dyslexia wanting to read efficiently using reading mode on the website, it appears obvious that users would want to stay in reading mode since that is theoretically providing the best conditions for them to read while they are on the website. This makes navigation an important part of assessing the performance of these tools from the human and context standpoint. As such DQH2 navigation and wayfinding is useful for targeting problems that specifically have to do with navigation.

5.2.5 Content structure and hierarchy

Headings are important on websites because they allow users with or without AT to sift through content and be aware of its hierarchy and relation to other webpage elements. Because reading mode transforms and renders web content anew to users [35], it is important that element structures are carried over to the reading mode interface. This was found not to always be the case in the evaluations. Text was found to be converted into italics, headings or text missing and creating gaps and stacked headings without content, and even text appearing that was not present in the website view. These problems were more difficult to assess using N10 because the evaluator had to combine different heuristics, such as N10H1 visibility of systems status and N10H6 recognition rather than recall or N10H4 consistency and standards, and argument for their interpreted violation to describe the problem. This process was more straightforward with the DQ heuristics and as such, for reading support technology that handles structured content, a heuristic for the specific purpose of navigation problems might be necessary.

5.2.6 Efficiency

Two of the N10 heuristics, N10H3 User control and freedom and N10H7 Flexibility and efficiency of use are on a surface level concerned with ease of access and user autonomy. When reviewing problems, many assessments had to be reconsidered where N10H3 had been incorrectly used to describe problems having to do with efficient access to reading mode and problems with freedom to choose how to view content, because the heuristic seem to represent user autonomy (control and freedom). This was an error on the evaluator's part coming from not reading the heuristic description properly. However, this also illustrates how heuristics can be misinterpreted based on previous experience and knowledge. This reaffirms the importance of preparation before heuristic evaluations but also highlights that the name and description of a heuristic is important to the method, exemplified by N10H3 User control and freedom is supposed to mean that users should have a clear way of leaving an unwanted interaction [45].

5.3 Types of problems found

After reviewing and categorizing the problems found in the evaluations, it could be discerned that the types of problems that were encountered using the different reading modes for the different reading tasks on the Act Church of Sweden website belonged to four main categories (see table 3): *Information loss upon rendering of content in reading mode, Limitations of navigation, Tool availability* and *Helpful features missing or working poorly*. Each of these will be presented in the sections that follow.

5.3.1 Information loss upon rendering of content in reading mode

The types of problems that have to do with rendering of content were discovered the most in the evaluations. Using Nielsens' set of heuristics, 34 problems could be found across all browsers and tasks and of these problems 6 unique problems were found. Using the Deque systems heuristics, 42 problems belonging to 12 unique problems were discovered. Safari had the least of these problems, but it is believed to be because the reading mode was only available on 7(41%) out of the 17 pages visited when evaluating Safari, and reading mode was available on all pages of Chrome and Firefox. Despite this, all pages where reading mode was available had lost part of their textual content. As found in the study by Li et al [35], Firefox Reader View algorithmically defines what content

would be suited for rendering, and that seems to be the case with Chrome and Safari as well. So, it is not unexpected that for example navigation menus, page footer content and breadcrumbs are missing. However, the evaluations found that the texts being rendered by reading modes would have a cutoff point even if the webpage equivalent contained more text like the rest (see figure 3).

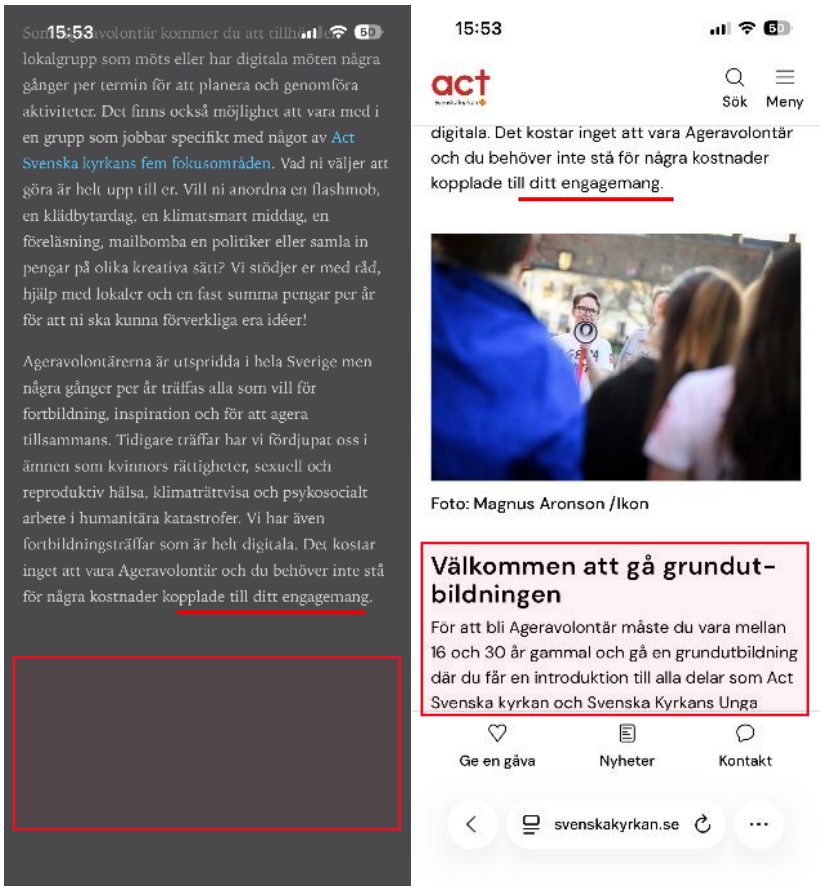


Figure 3: Reading mode textual content is cut-off's when there is more text on the webpage

Content seems to be rendered most reliably on blog post pages, such as pages *Act inleder almedagsveckan 2026* and *fasteaktionen 2026* [...], which is brought up along with articles as especially well suited pages by Mozilla Support [43] and findings in Li et al [35].

Other differences in content rendering was text that was being rendered by the Chrome reading mode in italics but appearing as regular text on the webpage, images which are replaced with placeholder boxes in Firefox reading mode, SVG symbols being rendered as images but not JPGs or PNGs in Firefox and loss of page structure and element hierarchy due to missing elements and/or images across all reading modes.

5.3.2 Limitations of navigation

Problems having to do with navigation rendered the majority of all severity scorings on the catastrophical level as per Nielsens definition [46]. These problems came about from attempting to navigate using links on pages within reading mode exclusively in the Firefox reading mode. Tapping a link takes users to a 404 error page on the Church of Sweden website (see figure 4) on 7 out of 13 links on the page "det här gör vi", 4 out 7 links on "svenska kyrkan inleder […], 3 out of 5 links on "fasteaktionen 2026 […]" and on all links accessible on pages "om Svenska kyrkan" and "organization och styrning".



Figure 4: 404 error page when navigating from within Firefox reading mode with.

These errors were shown to only appear in the reading mode, while links on the website itself worked. Although reading mode is supposed to support reading, links appearing in the rendered content misleadingly tell users that they may navigate from inside this view. Seeing this from a HAAT perspective of the *activity* being about reading on a website, it makes sense to allow users to stay inside reading mode when navigating between pages where reading mode is available, since this is the optimal way for them to read in the context of the website and tool. However, the current situation is that any navigation happening, in any of the browsers, takes the user out of reading mode, meaning that the user will have to into reading mode again before resuming reading. Altogether it can surmount to making the activity of reading tedious and frustrating to the users, and this makes easy access to reading mode more important.

5.3.3 Tool availability

The category of Tool availability included problems dealing with easily accessing the reading mode feature in the different browsers. Each of them had different ways of accessing it, with Safari and Firefox providing access directly through the address bar and Chrome through the browser menu. The severity of these problems may be rated low as a freestanding issue but becomes exacerbated when page navigation and time-based access is added to the context. Because user have the easiest time reading in reading mode but can't navigate from it, users have to resort to enabling and disabling it several times to for the reading and assimilation of knowledge throughout multiple webpages to lead to action and subsequently convert the reading activity to convert to becoming an engaged member in the Ageravolontär community. However, as it is, further studies are required to find out how severe the barrier of tool availability is for users.

5.3.4 Helpful features missing or working poorly

As research have established, there is a number of features such as read aloud, text highlighting and text adjustments that are known to be helpful to users with dyslexia who are reading web content. Many of these features are missing in the evaluated reading mode, most notably that reading mode rarely activates (on visited pages), read-aloud is available (only on Safari), text read-aloud being highlighted to accompany read-aloud, adjusting text taking users out of reading mode settings. Reflecting on the heuristic evaluations using the two sets, although none of them have heuristics regarding the presence of certain features, it was found to be easier to assign a heuristic to these kinds of problems using the DQ set. It is believed to be part because this set *prompts* evaluators

to look at and consider interaction methods (DQH1), navigation (DQH2) and visual and auditory alternatives (DQH10) that would otherwise, using the N10 set, be a matter of interpretation or having previous experiences of accessibility work. This is a known struggle in heuristics evaluations that is usually amended by involving multiple evaluators with different backgrounds, by establishing heuristics meaning prior to evaluations [63] or by constraining the scope of evaluation [12].

6 DISCUSSION

Through a heuristic evaluation of three different browser reading modes using two different sets of heuristics, this study has found that both heuristic sets can be used to find usability and accessibility problems in reading mode. There are however some differences between them and how they allow evaluators to view the problems from the standpoint of importance for the user and the organization. Furthermore, to answer the secondary research question, the HAAT model is believed to be an addition to the method of heuristic evaluation because it provides a framework for which the evaluator to look at the use of reading mode from the perspective of the human with a disability, the activity they should accomplish, the assistive technology itself and the context where the interaction is taking place. These matters will be discussed in the following sections.

6.1 Findings discussion

6.1.1 Heuristics usefulness for evaluating performance of reading mode

Both heuristic sets were found to be useful for finding usability and accessibility issues relating to the performance of reading mode used on the website of Act Church of Sweden for reading about three of its core information points. Using the N10 set, a total of 116 problems were found out of which 34 were unique, and using the DQ set a total of 90 problems were found, out of which 41 were unique. Upon analyzing the problems found in the heuristic evaluation, and subsequently comparing them between the heuristic sets, four main categories of problems could be discerned: *information loss upon rendering of content, limited navigation, tool availability* and *helpful features missing or working poorly*. Additionally, heuristics that were most useful for explaining these problems were those related to: *Consistency, use of language and readability, encountering errors, navigation, content structure and hierarchy* as well as *efficiency.* Looking at these results, we may see connections between the problems and heuristics that can inform us how to evaluate the performance of reading mode. **First**, rendering of content and the heuristics of consistency, readability, content structure and hierarchy appear connected, meaning that a performance factor of reading mode is that content that is transformed by the tool is done so accurately to avoid readability issues, and that it is done in such a way that is predictable. **Second**, limited navigation appears connected with heuristics of errors, navigation and efficiency, meaning that a performance factor of reading mode is that users can efficiently use the tool to both read and navigate to new pages to read on without encountering 404 errors. **Third**, tool availability appears connected with efficiency, meaning that a factor for reading mode performance is to give the user easy access to it. **Fourth**, features missing appear connected to readability and efficiency, meaning that another factor for reading mode performance is that a selection of features are available to the user to choose or adjust the settings that are best to support the users' individual preferences for reading. However, these factors can be further discussed in relation to their apparent severity.

The severity gradings issued for the categories of problems show that these problems were mostly minor problems, with the exception being that those graded with DQ received a grading bordering a major problem more than those graded with N10. Reflecting on this, it could mean that the DQ heuristics themselves cause evaluators to assess violations of these heuristics as more severe because of their tie to WCAG which is regarded as fundamental and important to the guarantee of good web accessibility according to laws and policies in the EU and Sweden. N10 on the other hand holds no such weight in accessibility domains, despite literature claims of an overlap between N10 and WCAG [15, 42], but can be useful for identifying problems having to do with fundamental usability issues such as clearly marked exits from reading mode or interface clutter that blocks modals. What is missing is thus believed to be the intent to evaluate accessibility, that can't be stumbled upon by following the N10 heuristics. Relating to this discussion of the needs of accessibility practice in the industry, such intents would have to be clarified in advance and furthermore to look at single disabilities at a time instead of all aspects of accessibility simultaneously, as the DQ heuristics attempts. Research on the other hand does not suffer from this decision-making because it already investigates one technology at a time, which is the case for this study's effort to evaluate performance for reading mode as part of reading support- and assistive

technology. For further research in methodology for evaluating performance of such tools through heuristic evaluations, researchers are recommended to choose a set of heuristics that best fit the intent and scope as well as the target user group [10, 41, 63].

6.1.2 Using HAAT to review problems

The human activity assistive technology (HAAT) model were used in this study to review the problems found in the evaluations similar to how a group review is usually done by a team of evaluators [63]. This allowed the evaluator to critically assess the severity of found problems based on what a user with dyslexia would need to efficiently read content on the website using reading mode. However, as HAAT is used in occupational theory, it's meant to help finding the best fit AT for the user [17]. In this study, it meant being strict in what features like read-aloud *should exist* or that navigation and access to the tool *should be easy* and is believed to be the main reason for these types of problems having a higher severity rating, which is reflected in severity ratings bordering on being major problems (see table 3). On the other hand, it would also mean lowering the severity grade of problems by one step or declaring it a non-issue due to the found problems not being critical to helping users read. Furthermore, during this review it became apparent, pertaining to the discussion about reading mode not being traditionally regarded as an assistive technology, that the assessment of reading mode as a tool to be strictly used by people with dyslexia had to be balanced with the overall function of the web browser. After all, reading mode exists in most web browsers and they can't cater to a specific user group but must be accessible for all. So having an easy access button to the tool can't take too much attention in the interface, which is for example believed to be a reason for the tool being placed in the menus over being placed in the address bar. To answer this study's secondary research question, this reasoning contributes to the conclusion that using HAAT in this study to determine AT best fit for a person with dyslexia reading on Act Church of Sweden was not a good fit for use in evaluating reading support technology, or mainstream user agents, such as reading mode. This would however not mean that it can't be used successfully to assess performance of other reading support technologies that are specifically built to support reading on websites for people with dyslexia. In regard to designers getting knowledgeable about accessibility, the use of the model in this study underscores that HAAT in itself cannot supply designers with knowledge about impairment, the activities to be carried out, the AT nor the context of its use. But writing a brief outlining all components of HAAT present in the target of evaluation can complement designers knowledge about usability in AT by providing a framework by which to investigate and prompt thinking about the needs of people with impairments, starting from which activity they need to support. For future uses involving HAAT and heuristic evaluation, researchers and industry professionals alike are urged to involve users or domain experts [2, 7, 10, 60, 63] to obtain a more accurate understanding of users with disabilities such as dyslexia.

6.1.3 Study limitations

The previous experience of the evaluator led to initially misinterpreting heuristics and confusing for example control and freedom with autonomy, and match between system and real world with match between reading mode and the website. It is not an unusual phenomenon [63] but one that is undesired when it comes to finding and diagnosing real problems. However, such misinterpretations of heuristics and guidelines can have consequences such as focusing development resources on problems that are not real problems, to the detriment to the user experience. Had this been a government agency or an e-health platform instead of a charity organization, it would have resulted in very real issues for the well-being of people with disabilities [33]. This also highlights the importance of testing heuristics that are to be used, as two sets in this study were.

Although this study found that the two heuristic sets are useful for heuristic evaluation, it is important to note that these findings are based on the results from a single evaluator and that entails the results come with a drastically lower reliability than if multiple evaluators were included [46]. Furthermore, having one person carrying out all evaluations makes them prone to learning effects, in this case meaning knowledge of usability problems would carry over between tests because the evaluator would remember the problems between the evaluations using the two heuristic sets. On the one hand this makes it hard to prove that the Deque set, which was used after the Nielsen set, could truly find both usability and accessibility problems, but on the other hand it shows how different heuristics can be used to accurately diagnose found problems, which Deque set did better for some of the main problem categories.

This study has only evaluated the performance of reading based on one website, and the results are thus not generalizable to reading mode use on other websites. Although the study by Li et al [35] corroborates the found problem category of *Information loss upon rendering of content in reading mode*, problems relating to navigation and feature availability are insofar only discovered

on the website for Act Church of Sweden and it is unknown if errors with navigation is a symptom of coding practice or reading mode interpretation.

6.2 Heuristics for evaluation of reading mode

The results from the heuristic evaluation pointed out a lack of heuristics that could accurately describe the main categories of problems. First, the Nielsen (N10) and Deque systems (DQ) set of heuristics were shown to have different uses for explaining different kinds of problems. Notably the DQ heuristics provided more clarity and were easier to understand and subsequently use to diagnose problems related to performance. In comparison, Nielsens heuristics were broadly interpreted which made them difficult to use for accurately diagnosing problems, even though they theoretically can be used to evaluate accessibility [15, 42]. This speaks to domain specific heuristics being more useful than a general set of heuristics. As such, a contribution to research in reading support technologies are here presented in the form of six new- and four adjusted heuristics which are based on study findings in chapters 5.2 and 5.3 and corroborated by both Nielsen and Deque systems set of heuristics. This heuristic set thus is part of heuristics developed using mixed methods: literature review, study findings and previous heuristics [52]. The heuristics for *interaction methods and modalities, consistency and predictability, help and documentation and prevent and communicate errors* are largely versions of the N10 and DQ heuristics combined and will not be covered here, and are instead presented here in short (see table 6), the full version is found in appendix D. The remaining six new heuristics are described below.

Table 6: New and adjusted heuristics for reading mode evaluation

Heuristic	Status	N10	DQ
Preserve and enhance content structure	New		X
Provide easy navigation	New		X
Use plain language	New	X	X
Allow efficient tool access	New	X	
Allow readability adjustments	New		X
Provide user feedback	New	X	
Interaction methods and modalities	Adjusted		X
Consistency and predictability	Adjusted	X	X
Help and documentation	Preserved	X	
Prevent and communicate errors	Adjusted	X	X

6.2.1 Preserve and enhance content structure

The Deque heuristics and WCAG hold that the structure of a website is important in a visual and programmatical sense. The structure of web content as rendered by reading mode should be given similar importance pertaining to the performance of the tool. Because this study has found that reading mode exhibits problems with rendering and preserving content and structure of Act Church of Sweden, future evaluations of reading mode or similar reading support technology that transforms website content should do so in a way that preserves this structure and enhances it so that users can easily read and not lose any information because of the reading support technology they choose to use.

6.2.2 Provide easy navigation

Informed by the Deque heuristics and review with HAAT, any tool that makes text easier to read by providing an alternative view should also preserve the navigation capabilities of a website. This makes it so users would not have to enter/exit out of the tool every time they wish to go to another page where an alternative view is available. Additionally, this heuristic should be useful for describing errors that may result from navigation, as was found in the heuristic evaluations.

6.2.3 Use plain language

Text that is written in plain language is easier to understand because it uses words that most people are familiar with and short sentence structures. Problems found relating to this heuristic were about the labels and descriptions of features such as forward/backward audio modals. Plain language is also referenced in ISO 24495-1:2023[32], is known to benefit everyone, and is referenced in the Deque set and by interpretation in the Nielsen set. While no problems relating to language use was documented on the website itself, reading support tools that, for example, provide synonyms or provide alternative interpretation of textual content should follow the same principle.

6.2.4 Allow efficient tool usage

Efficiency was seen to be referenced in two of the N10 heuristics, N10H3 User control and freedom and N10H7 Flexibility and efficiency of use but were not interpreted to sufficiently cover the needs of efficiency in the use of reading mode. This heuristic gives clarity to this, and frames efficiency as making it easy for users to see, reach and enable/disable reading support technologies from any part of the interface. This includes N10H3 notion of clearly marked exits and N10H7 notion of shortcuts.

6.2.5 Allow readability adjustments

Readability is widely referenced as an important factor to the activity of reading in literature about dyslexia and interface design. Literature about dyslexia and technology use especially highlights that users should be able to freely adjust typography according to preference because dyslexia is experienced in varying severity. Additionally, typographical principles relating to content adjustments are also mentioned in the WCAG [66] Perceivable and Operable principles, for example 1.4.4 Resize text (AA) and 2.4.6 Headings and Labels (AA) and by extension in DQH6 Language and readability and DQH5 Contrast and legibility. Furthermore, reading mode has shown to provide options for color themes and fonts in varying ranges (See Appendixes G, H & I) that can be improved to cover more fonts. This motivates the necessity of a heuristic that covers readability and the adjustments of typographical factors.

6.2.6 Provide user feedback

Where read-aloud features are available, they should provide users with feedback that help them understand what is happening in the interface. This is especially based on the found problems with audio modal in Safari reading mode, which didn't provide textual or audio feedback on why audio couldn't be replayed, or how the forward/back audio modals worked. Additionally, the lack of consistency of reading mode availability and capability to predictably render content should be reflected in the reading mode interface so users can manage expectations of its usefulness, for example by showing "60% of the page was rendered" or otherwise communicate when and what content isn't shown to users. Such status updates stem from N10H1 Visibility of system status but are here contextualized based on the user with dyslexia using reading mode.

7 CONCLUSION

Digital accessibility is important to research and practice because people with disabilities such as dyslexia will otherwise risk becoming excluded from physical, cultural and digital environments that contribute to wellbeing, communication and community building in society. To contribute to making digital environments more accessible, this study has answered the call for research into a methodology for evaluation the performance of reading support technology that is used by people with dyslexia [3]. It is also motivated by the dearth of knowledge in the practice of digital accessibility among UX professionals in industry settings. The target of study has been reading mode, a reading support- and assistive technology that is available on almost every web browser and which groups a number of features that are known to help people with dyslexia to read. To evaluate the performance of reading mode the method of heuristic evaluation is proposed as a suitable method to identify usability and accessibility problems that require less resources than traditional user testing. Two sets of heuristics, Nielsens 10 usability heuristics and Deque Systems accessibility heuristics, were tested in evaluations of three different reading modes to find out the extent to which they can be used to evaluate reading mode in the case of the website for Act Church of Sweden, that strives to know more about the use of assistive technologies on their website. The HAAT model was applied in the problem reviewing process. A brief was written, covering all components in the model, and was then used to judge problem severities. This was found to be beneficial in ruling out non-issues but also made these rulings quite strict on the basis of the AT being supposed to be a "best fit" for the user. The evaluations found that while both

sets could be used to find the same types of problems, the Deque systems set was more specific to explain the type of problems that were encountered. The study concluded with a contribution to research in the form of a set of heuristics that can be used to evaluate the performance of reading mode and possibly other reading support technologies in the future, towards the goal on making websites accessible to all readers.

Future work

This study has contributed to methodological research in evaluating performance of reading mode for people with dyslexia, but this is only one example of such a methodology. Because the results show that problems are able to be found using this method, future work related to this study should consider using heuristic evaluation for other reading support technologies as well. If the proposed heuristics should be used, they need to be tested sufficiently, for example by investigating if evaluator can use them to find real usability and accessibility problems, as Nielsen and Molich [47] did by establishing problems and seeing how many of them different evaluators can find. However, to find out what constitutes real problems for people with dyslexia using reading mode, user studies or participatory evaluations should first be carried out and be compared to the findings in this study. In regard to the use of the HAAT model, it is believed that it may be of use in the field of HCI to build a comprehensive understanding of AT use for people with disabilities. As such it is suggested that researchers continue to explore its use to inform study design [24] or to review usability issues relating to AT performance, though it would be beneficial to let experts or users inform the mapping of its components to the relevant situation.

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Appendix A: Heuristic evaluation tasks

Sites mentioned in the tasks are translated to English for ease of understanding.

Task 1: Reading about- and making a one-time donation for "fred och demokrati/Peace and democracy"

1. Go to the page "Ge en gåva/Donate" and then to the page "Stöd ett fokusområde/Support an area of focus". Check reading mode availability and features.
2. Read about each of the focus areas and what donated money would go to. Then go to the page for "Fred och Demokrati/Peace and democracy". Use reading mode whenever possible.
3. Fill out information in the form for making a one-time donation through Swish, go to accept then cancel payment. Read the information on the page. Use reading mode whenever possible.

Task 2: Reading about the cause of ACT Church of Sweden.

1. Go to the page "Det här gör vi" and browse through the page to form a first impression of the work that Act Church of sweden does.
2. Go the page "Om Act Svenska kyrkan", read the contents of the page using reading mode.
3. Go the page "Organisation och styrning", read the contents of the page using reading mode.

Task 3: Reading about how to become an engaged member in ACT Church of Sweden.

1. Go to the page "Engagera dig" and browse through the page to see possible ways of becoming engaged in the cause. Check reading mode availability on pages you visit, look at how the features of reading mode work with the page elements.
    a. Scroll to "Nyheter för engagerade i Act Svenska kyrkan" and into the post "Svenska kyrkan inleder Almedalsveckan 2026". Read the contents of the page using reading mode. Then go back and to "Fasteaktionen 2026: 35 miljoner kronor från våra händer till där det händer". Read the contents of the page using reading mode.
2. Go to the page "Ageravolontär". Browse the contents of the page with and without reading mode.
    a. Read the contents under the heading "Mer om Agera I församling".

Appendix B: Heuristic evaluation template

Task X							
ID	**Problem**	**Location**	**Heuristic**	**Severity**	**Reasoning**	**Images**	**Images**
Unique problem ID	*Short problem title*	*Where/when*	*4 Consistency and standards*	*2*	*Longer problem description*	*Image showing described problem*	

Appendix C: Evaluation results

Set name	Nielsen's 10	Deque systems
Authors	Nielsen	Boudreau & Pasi
Date	1994	2019
Heuristic 1	Visibility of System Status	Interaction methods and modalities
Heuristic 2	Match Between the System and the Real World	Navigation and wayfinding
Heuristic 3	User Control and Freedom	Structure and semantics
Heuristic 4	Consistency and Standards	Error prevention and states
Heuristic 5	Error Prevention	Contrast and legibility
Heuristic 6	Recognition Rather than Recall	Language and readability
Heuristic 7	Flexibility and Efficiency of Use	Predictability and consistency
Heuristic 8	Aesthetic and Minimalist Design	Timing and preservation
Heuristic 9	Help Users Recognize, Diagnose, and Recover from Errors	Movement and flashing
Heuristic 10	Help and Documentation	Visual and auditory alternatives
Method of documentation	ID. Problem, Heuristic, Severity, Reasoning	ID, Problem, Heuristic, Severity, Reasoning
Added method of documentation	Location (Where/When)	Location (Where/When)
Total problems found	116	90
Total unique problems	35 (34)	41 (38)
Unique problems declared non-issues	1	3
Overall most mentioned Heuristic of total problems	H4 (77)	H7 (61)
Overall least mentioned Heuristic of total problems	H10 (3)	H10 (1)
Overall mean severity of total problems	1,86	2,28
Main problem categories		
Content rendering		
Total problems found	34	42
Unique problems found	9	12
Mean severity	1,7	2,2
Limited navigation		
Total problems found	15	10
Unique problems found	6	7
Mean severity	2,1	2,9

Tool availability		
Total problems found	12	13
Unique problems found	6	9
Mean severity	1,8	2,0
Missing or poorly working features		
Total problems found	18	12
Unique problems found	11	8
Mean severity	2,8	2,6
Lesser problem categories		
Visually communicating function		
Total problems found	2	2
Unique problems found	1	2
Lacking notifications and feedback		
Total problems found	3	0
Unique problems found	3	0
Problems requiring code inspection		
Total problems found	3	3
Unique problems found	3	3
Safari total problems found	37	28
Unique problems found	18	18
Mean severity	2,1	2,1
Task 1	16	11
Task 2	8	11
Task 3	13	6
Chrome total problems found	37	25
Unique problems found	15	12
Mean severity	1,9	2,1
Task 1	21	8
Task 2	7	6
Task 3	9	11
Firefox total problems found	42	37
Unique problems found	19	18
Mean severity	1,8	2,6

Appendix D: Proposed heuristics

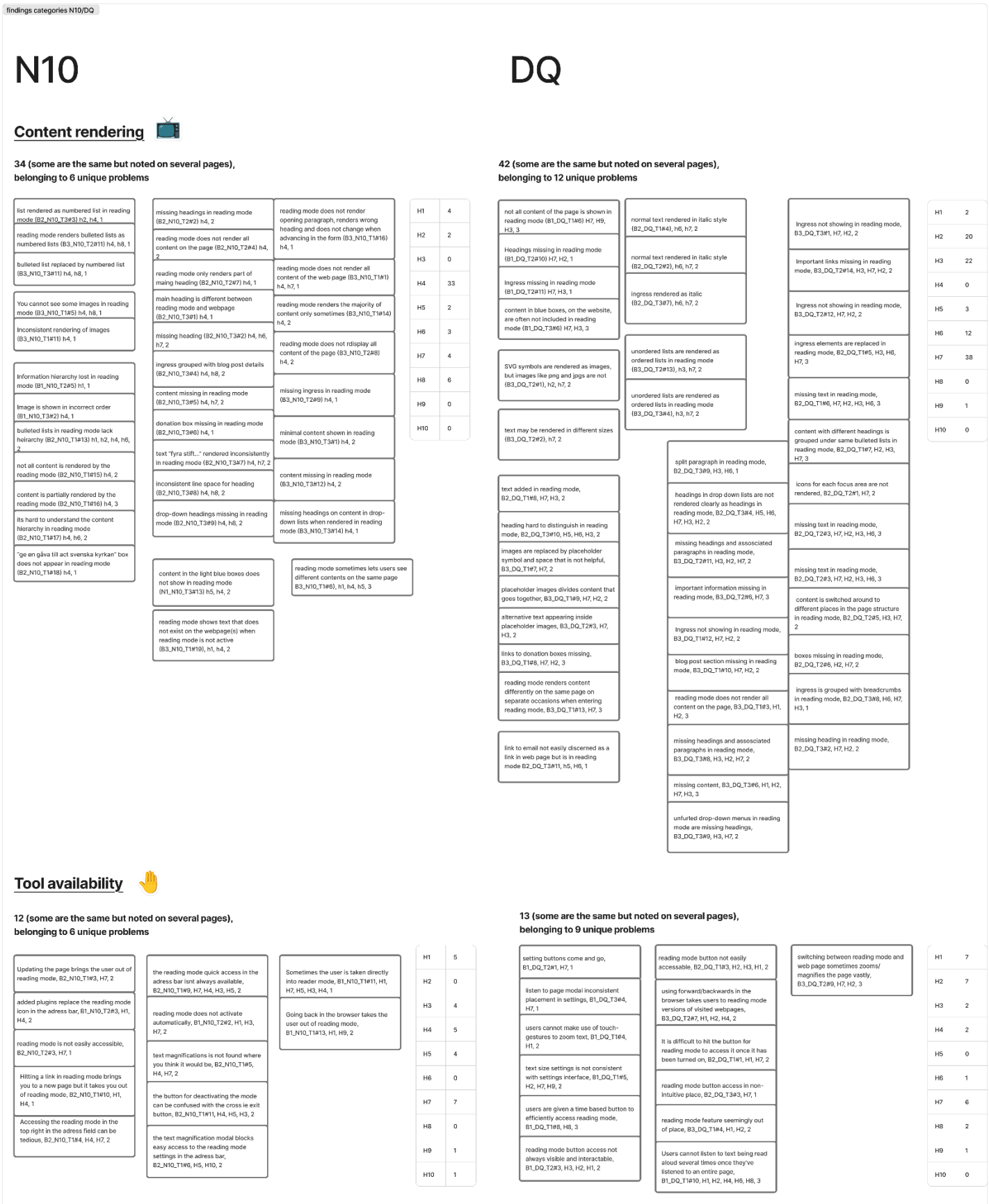
Heuristic	Description	Status	N10	DQ	WCAG Percievable	WCAG Operable	WCAG Understandable	WCAG Robust
Preserve and enhance content structure	Content that is transformed by the assistive technology preserves its structure. Users can make sense of the structure of the content.	New		H3 Structure and semantics	X		X	
Provide easy navigation	Users can easily navigate and find content from within and outsite the assistive technology.	New		H2 Navigation and wayfinding		X	X	
Use plain language	Language that is used in content and to explain functionality is written is such a way that it can be understood by anyone.	New	H2 Match between the system and the real world	H6 Language and readability	X	X	X	X
Allow efficient tool access	Users can easily see and reach/enable/disable the assistive technology from any part of the interface.	New	H3 User control and freedom, H7 Flexibility and efficiency of use		X	X		
Allow readability adjustments	Users can make adjustments to text so that it becomes easier to read.	New		H6 Language and readability, H5 Contrast and legibility	X		X	X
Provide user feedback	Users actions should have percievable feedback that is easy to understand.	New	H1 Visibility of system status		X	X		X
Interaction methods and modalities	Users can efficiently interact with the system using the input and output method of their choosing.	Adjusted		H1 Interaction methods and modalities	X	X		X
Consistency and predictability	The interface follows usability standards. Interaction element are used consistently throughout the interaction in a manner that is predictable to the user.	Adjusted	H4 Consistency and standards	H7 Predictability and consistency	X	X	X	
Help and documentation	It is easy to reach help and documentation that explains functionality to the users, but its best if its unnessecary.	Preserved	H10 Help and documentation			X	X	X
Prevent and communicate errors	Errors should be prevented, and should they arise they are clearly communicated to the user and suggests what to do next.	Adjusted	H5 Error prevention, H9 Help users recognize, diagnose and recover from errors	H5 Error prevention and states	X		X	X

Appendix E: Problems organized by severity

#	Safari N10	ID's	Safari N10 Av severity	Safari DQ	ID's	Safari DQ Av severity	Chrome N10	ID's	Chrome N10 Av severity	Chrome DQ	ID's	Chrome DQ Av severity	Firefox N10	ID's	Firefox N10 Av severity	Firefox DQ	ID's	Firefox DQ Av severity	Unique problems across tools N10 (across browsers)	ID's	Unique N10 Average severity	Unique problems across tools DEQUE	ID's	Unique DQ Average severity
1	Loss of information hierarchy in reading mode ()			Links not visually communicating their function as within page links ()			Limited navigation possibilities from reading mode ()			email link clickable in reading mode but not on webpage, email links seem more visually clickable in webpage view ()			Reading mode features for adding to reading list are not visually or textually described to the user ()			Reading mode does not render all content on the webpage, important information or content parts may not be seen by users ()			Inconsistency in tool menu items ()			Links not visually communicating their function as within page links ()		
2	Reading mode way of rendering website content (what scroll position you land on())			Setting buttons have inconsistent placements ()			Users are not notified of reading mode availability ()			Difficult to physically tap the reading mode icon in across bar ()			Images are not rendered in a useful way in reading mode, they have placeholders ()			Reading mode features list seems out of place amongst reading list features ()			Users are not notified of reading mode availability ()			email link clickable in reading mode but not on webpage, email links seem more visually clickable in webpage		
3	Settings: menu items inconsistent ()			I can be hard to distinguish links in sepia theme ()			Users are not made aware of how reading mode renders content ()			Within-page navigation disappears in reading mode ()			Text is coded as an external descriptive text for images () ie. not alt text but a paragraph(), and is rendered in reading mode but not to image. Same goes for photographer credits ()			Images are represented in reading mode with placeholders and sometimes alt text that is not particularly useful to reduce distractions ()			Users are not made aware of how reading mode renders content ()			Setting buttons have inconsistent placements ()		
4	Efficiently reaching tool, activating ()		2	Heading contents are different between reading mode and webpages ()		2	Bulleted list is rendered as numbered list in reading mode ()		1	Reading mode is not easily accessible ()		2	reading mode renders bulleted lists as numbered lists ()		1	SVG symbols are rendered as images, but images like png and .jpgs are not ()		2	Bulleted list is rendered as numbered list in reading mode ()		1	I can be hard to distinguish links in sepia theme ()		1
5	Extra interaction steps in reading mode ()		2	Interacting with text magnification modes takes you out of the settings interface ()		2	Access to reading mode is not efficient ()		2	Regular text in reading mode may be rendered in italic style ()		2	Skipping heading levels from h1 to h3 impacts heading design in reading mode ()		1	Text may be rendered in different sizes ()		2	Images are not rendered in a useful way in reading mode, they have placeholders ()		1	Tool availability ()		2
6	Feature explanations lacking ()		2	Tool availability		2	Updating, going back and going forwards on pages takes users out of reading mode ()		2	Content is rendered so that sections of text or headings are out of place or sometimes it is not rendered at all ()		2	Reading mode does not render all content on the web page ()		2	Going backwards/forwards in the browser takes can sometimes take users to reading mode versions of visited webpages ()		2	Skipping heading levels from h1 to h3 impacts heading design in reading mode ()		1	Read aloud only works when reading mode does ()		2
7	Feature symbol communicating function ingrossly ()		2	Read aloud only works when reading mode does ()		2	Access to text magnification is not efficient ()		2	Reading mode renders text that seemingly does not exist anywhere on the website ()		2	Users can only choose from two fonts in reading mode ()		2	Browser history shows duplicates of the same webpages and does not specify which one was visited with reading mode ()		2	Alt-text sometimes bleeds into reading mode movie image placeholders. It hinds little meaning since images are not displayed to reduce distractions ()		2	Reading mode: Cannot magnify text (zoom out/in) with gestures ()		2
8	Limited navigation capability ()		2	Reading mode: Cannot magnify text (zoom out/in) with gestures ()		2	Text magnification mode blocking access to other features in the adress bar ()		2	The browser navigation (backwards, forwards), takes users out of reading mode ()		2	Reading mode may render a different heading than on the rendered webpage ()		2	Reading mode only has two fonts to pick from ()		2	Reading mode may render a different heading than on the rendered webpage ()		2	Inconsistent placement of text magnification UI ()		2
9	Missing content ()		2	Inconsistent placement of text magnification UI ()		2	Users can be confused on how to exit the mode()		2	Some headings are different in reading mode as compared to the webpage ()		2	Alt-text sometimes bleeds into reading mode movie image placeholders. It hinds little meaning since images are not displayed to reduce distractions ()		2	Bulleted list are sometimes rendered as numbered lists ()		2	Going backwards in the browser from reading mode does not act how you would expect, it usually disables reading mode instead of going back to the previous page, it means that reading mode is counted as a separate web page ()		2	Missing content ()		2
10	Plugin integration conflict ()		2	Missing content ()		2	Content is rendered differently from the webpage in reading mode, often se that content and context is lost ()		2	Reading mode may render headings in a way that makes them harder to read or distinguish on a page ()		2	There is no smooth transition between reading mode and webpage, it may be needed because the nature of the reading mode is that you cant easily navigate from within it, you have to switch views ()		2	The tool does not have a read aloud feature ()		3	Tool is soldomly available ()		2	Access to reading mode is not efficient ()		2
11	Read aloud content is prone-assisted incorrectly ()		2	Very limited navigation from inside reading mode ()		2	Reading mode renders content that do not exist on the webpage, often headings that are wanted differently. Extra paragraphs appear in some condition pages ()		2	Switching between webpage and reading mode leaves users in non-consistent scroll position that may confuse users ()		3	Important information such as GDPR is consistently left out of reading mode ()		2	Readspeaker access button "lyssna" shows in reading mode but take users out of the site ()		3	Limited choice of fonts ()		2	SVG symbols are rendered as images but images like png and .jpgs are not ()		2
12	Reading mode is an inaccurate mirror of website content ()		2	Long and unnecessary animations upon hiding elements ()		2	Users are not notified of missing information (content) in reading mode ()		2	Reading mode has no read aloud feature ()		3	SVG images are rendered in full inside reading mode, even though for example png and .jpg are not ()		2	Users sometimes have none or few options to navigate from within reading mode ()		3	Efficiently reaching tool, activating ()		2	Text may be rendered in different sizes ()		2
13	Switching between page and reading mode views efficiently (where in terms of scroll position does users land when switching) ()		2	Users can hide elements but they are still read aloud ()		3	Features that can be combined into one tool are split into different menus ()		3				The "lyssna" readspeaker plays content that cannot be seen in reading mode ()		2	Headings are different in reading mode as compared to website view ()		3	Limited navigation capability ()		2	Going backwards/forwards in the browser takes can sometimes take users to reading mode version of visited webpages ()		2
14	Tool availability (seldom available)		2	Reversing hidden elements is not efficient ()		3	Content cannot be read aloud ()		3				The "lyssna" readspeaker does not always appear ()		2	Links to direction boxes are missing in reading mode, users are missing this primary element if they are using reading mode ()		3	Feature symbol not communicating function properly ()		2	Browser history shows duplicates of the same webpages and does not specify which one was visited with reading mod		2
15	Read aloud availability (seldom available())		3	Access to reading mode is not efficient ()		3	When the "lyssna" readspeaker link takes users to a page only an audio media, users cant follow along with the text ()		4				Going backwards is the browser from reading mode does not act how you would expect, it usually disables reading mode instead of going back to the previous page, it means that reading mode is counted as a separate web page ()		2	Transitioning between reading mode and webpage is unpredictable and inconsistent when it comes to the scroll position that users land on ()		3	Loss of information hierarchy in reading mode, due to missing headings/ images/text ()		2	Reading mode only has two fonts to pick from ()		2
16	Read aloud: does not read all content on website, only reading mode content ()		3	No reading aloud tool highlighting ()		3							Reading mode is inconsistent in when it renders what content, this hard to predict ()		3	Reading mode may sometimes render content on the same page in a different way ()		3	When switching between web page and reading mode, users land on different scroll positions ()		2	Bulleted list are sometimes rendered as numbered list		2
17	Read aloud: no highlighting of text to read along with audio ()		3	Users cannot re-listen to audio after all of it has been played ()		3							Changing text size in reading mode sometimes affects the zoom on the webpage when exiting the mode ()		3	Switching between website and reading mode may sometimes zoom the website in wonky ()		3	Conflicts with plugin integration, change of symbol ()		2	Long and unnecessary animations when hiding stuff		2
18	Read aloud: cannot focus several times ()		4	Audio media not explaining back/ forward symbols with text ()		3							Activating the "lyssna" feature in reading mode takes the user to a new webpage with only an audio model that plays that page without visuals. ()		3	Some links takes users to 404 error sites on the church of sweden web sites ()		4	Read aloud: content is pronounced incorrectly ()		2	Interacting with text magnification modes takes you out of the settings interface ()		2

| | | |

Appendix F: Problems organized by category



Features not present or working poorly 🛠️

18 (some are the same but noted on several pages), belonging to 11 unique problems

- Reading mode is seldomy available, B1_N10_T1#1, H4, H1, 3
- There is no explanation of why reading mode does not activate on pages, B1_N10_T1#7, H10, 3
- text being read aloud is not spelled correctly, B1_N10_T1#13, H4, 2
- reading mode cannot be enabled, B1_N10_T2#1, H1, 2
- reading mode does not activate, B1_N10_T3#1, H7, 1
- activating "lyssna" inside of reading mode takes user out of the website, B3_N10_T1#2, H1, H2, H3, H4, H5, H6, 3
- you cannot easily follow along with the text being read aloud, B1_N10_T3#7, H7, H1, 2
- lyssna button is out of place, B3_N10_T3#4, H4, 2
- activating "lyssna" inside of reading mode takes user out of the website, B3_N10_T1#8, H1, H2, H3, H4, H5, H6, 3
- the lyssna feature brings users to a separate site, B3_N10_T3#5, H5, H9, H3, 3
- users cannot have content read aloud, B2_N10_T1#8, H4, H7, 3
- It is hard to follow along in the text while "listen to page" is active, B1_N10_T1#10, H1, H6, 3
- you cannot re-listen to the text being read aloud, B1_N10_T3#6, H1, H5, H9, 3
- no explanation nor feedback of why users cannot listen to text several times, B1_N10_T3#8, H5, H10, H7, H1, H9, 4
- You cannot re-listen to text being read aloud, B1_N10_T3#9, H1, H5, H9, 3
- Invisible text shown in reading mode, readiing aloud error, B2_N10_T1#20, H1, H5, H9, 4
- text being read aloud is based on what content is rendered in reading mode, not actual website content, B1_N10_T3#10, H5, H1, 3
- Features that might be of use to dyslexic users are separated into different menus, B2_N10_T1#7, H7, 3

H1	11
H2	2
H3	3
H4	6
H5	8
H6	3
H7	5
H8	0
H9	9
H10	1

12 (some are the same but noted on several pages), belonging to 8 unique problems

- Limited amount of fonts available, B3_DQ_T2#10, H5, H6, 2
- reading mode is not available, B1_DQ_T1#1, H6, H1, H5, 3
- reading mode not available, B1_DQ_T3#1, H6, H1, H5, 3
- safari only reads text to users on pages were reading mode is available, B1_DQ_T1#3, H6, H10, H7, 2
- links not acting as expected, B1_DQ_T1#2, H7, H9, H3, 1
- the "lyssna" button shows in reading mode but takes users out of the site, B3_DQ_T1#2, H7, 3
- the "lyssna" button shows in reading mode but takes users out of the site, B3_DQ_T3#3, H7, 3
- reading mode does not read content to user, B2_DQ_T3#6, H1, H6, 3
- tool does not have a read aloud feature, B3_DQ_T1#1, H1, H6, 3
- Users cannot listen to text being read aloud several times once they've listened to an entire page, B1_DQ_T1#10, H1, H2, H4, H6, H8, 3
- user cannot easily follow along with text that is being read aloud, B1_DQ_T1#9, H2, H3, H6, H5, H1, 3
- users being taken out of reading mode settings after adjusting text size, B1_DQ_T2#9, H8, H4, H1, 2

H1	7
H2	2
H3	2
H4	2
H5	4
H6	8
H7	4
H8	2
H9	1
H10	1

Limited navigation 

15 (some are the same but noted on several pages), belonging to 6 unique problems

- text magnifications is not found where you think it would be, B2_N10_T1#5, H4, H7, 2
- the text magnification modal blocks easy access to the reading mode settings in the adress bar, B2_N10_T1#6, H5, H10, 2
- when reading mode is initiated the main heading is barely visibile, B1_N10_T2#6, H1, 1
- enabling reading mode always takes the user to the top of the page no matter the scroll position on the webpage, B1_N10_T2#7, H5, H1, H4, 2
- disabling the reading mode always takes the user to the top of the page no matter the scroll position in reading mode, B1_N10_T2#8, H6, H1, H4, H3, 2
- the users are met with errors when navigating from pages inside reading mode, B3_N10_T1#10, H4, H5, H7, H9, 4
- some links show error 404 when attempting to navigate from inside reading mode, B3_N10_T2#7, H4, H5, H9, H3, 3
- some links show error 404 when attempting to navigate from inside reading mode, B3_N10_T3#7, H4, H5, H9, H3, 4
- the button for deactivating the mode can be confused with the cross ie exit button, B2_N10_T1#11, H4, H5, H3, 2
- Users cannot navigate the website when reading mode is active, B1_N10_T1#15, H1, H4, 2
- limited navigation from within reading mode, B1_N10_T3#4, H6, H7, 2
- Hitting a link in reading mode brings you to a new page but it takes you out of reading mode, B2_N10_T1#10, H1, H4, 1
- Users cannot navigate the entire website in reading mode, B2_N10_T1#21, H4, 1
- hitting back in the browser disables reading mode instead of going back a page, B3_N10_T3#9, H4, H7, 2
- enabling reading mode when the reader is far down the page always places the scroll position at the top, B1_N10_T3#3, H3, H4, 1

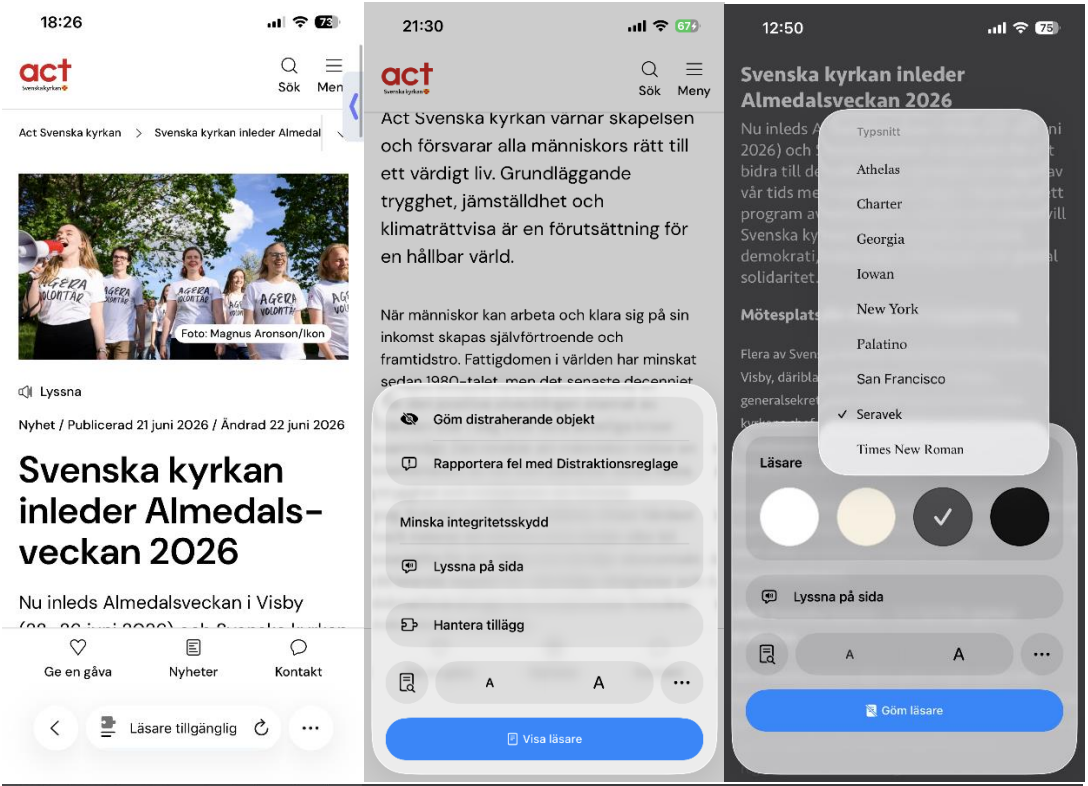
H1	3
H2	0
H3	5
H4	12
H5	6
H6	2
H7	4
H8	0
H9	3
H10	1

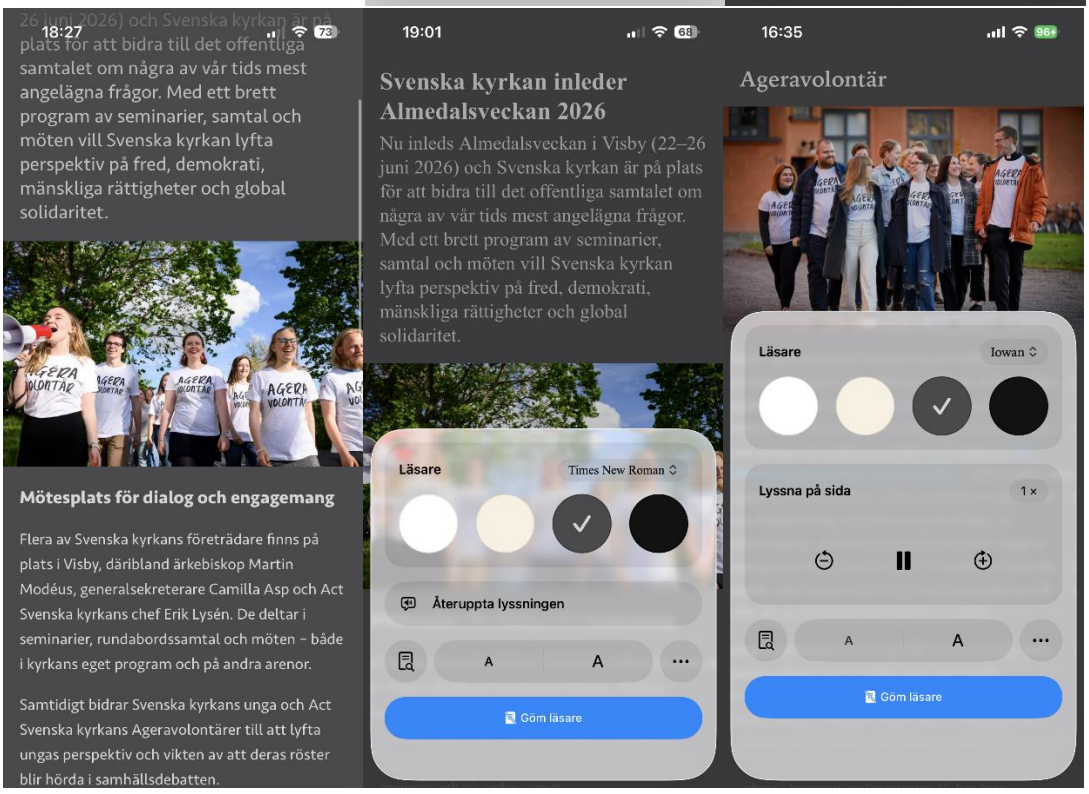
10 (some are the same but noted on several pages), belonging to 7 unique problems

- transition between reading mode and web page view is unpredictable, B3_DQ_T1#11, H7, H1, 3
- reading mode does not allow user to navigate the web page, B1_DQ_T1#7, H2, 2
- very limited options for navigation, B3_DQ_T3#5, H1, 3
- hitting back or forward in the browser takes user out of reading mode, B2_DQ_T2#4, H1, H7, 2
- elements of navigation dissapears in reading mode, B2_DQ_T1#2, H2, H7, 2
- forwards/backwards visited pages history does not clarify what pages are in reading mode, B3_DQ_T2#8, H2, H4, 2
- switching between reading mode and web page sometimes zooms/ magnifies the page vastly, B3_DQ_T2#9, H7, H2, 3
- links leading to 404 error pages, B3_DQ_T1#5, H4, 4
- hitting links inside reading mode takes user to 404 error site, B3_DQ_T2#4, H4, 4
- hitting links inside reading mode takes user to 404 error site, B3_DQ_T3#7, H4, 4

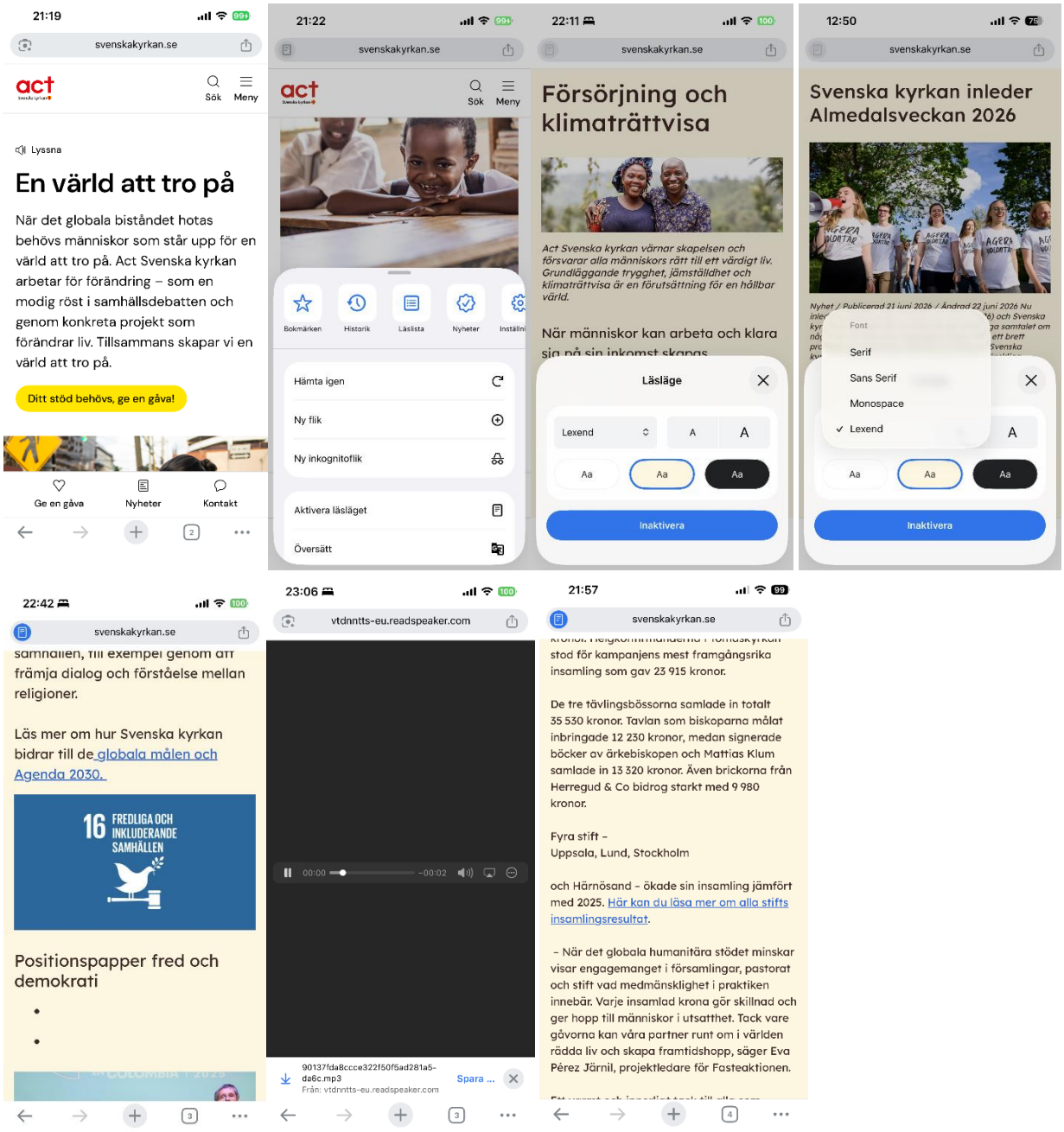
H1	1
H2	4
H3	0
H4	4
H5	0
H6	0
H7	4
H8	0
H9	0
H10	0

Appendix G: Screenshots safari reading mode



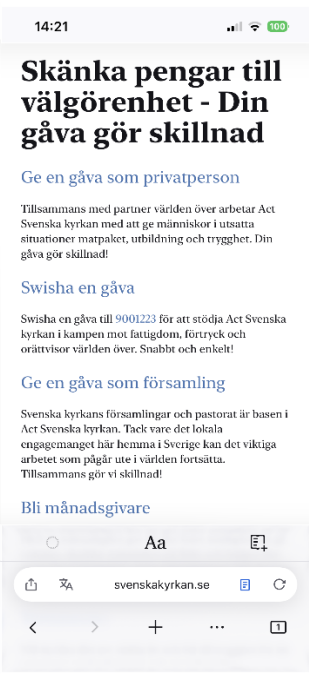


Appendix H: Screenshots Chrome reading mode



Appendix I: Screenshots Firefox reading mode

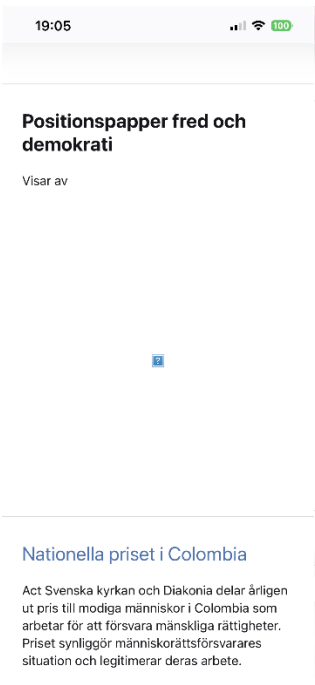
















Appendix J: HAAT brief for reviewing problem severities

In essence, this brief of these components to this particular interaction provides requirements for how reading mode would need to function to provide assistance with reading for people with dyslexia.

Human component

Person with dyslexia

Word reading
Word reading is about the ability to decode and recognize printed words have to do with processing orthography (word and letter shape) of text such as confusing p and q or b and d or words like addition and audition [1, 3, 6]. Difficulties pertaining to word reading is a primary characteristic of dyslexia but can vary in severity [3]. It may cause difficulties with oral reading and reading comprehension because of difficulties with recognizing words in a text. It may also affect reading accuracy and speed.

Spelling difficulties
It requires cognitive effort to retrieve the spelling of a word. And for people with dyslexia, it can overload the working memory, which causes problems for planning, idea formation and grammatical formulations in writing [3].

Reading comprehension
Reduced reading and writing experience resulting from inborn dyslexia can according to Catt et al [3] impact aspects of language such as the understanding and use of grammatical rules and discourse structures. The ability to develop reading skills is hindered by dyslexia.

Emotional factor
Not being able to read at a certain speed or with a certain understanding may impact self-perception and esteem as well as reading skills [5]. Anxiety related to dyslexia can be particularly severe in college, and in work environments experiences of dyslexia can negatively impact confidence [7].

Activity component

Reading - informed by readability [2]

Reading on the purpose axis
A reader will likely read for the purpose of content uptake to learn about processes such as how to donate or get involved in the community.

Reading on the time axis
In regard to time that they will do glanceable reading like skimming and deep reading to find and learn for example about how to become engaged in the community or to find out if Act Church of Sweden is the right organization to donate to.

AT component

Users may choose which tool works best for them given their preferences. The AT should help the person with dyslexia achieve their goal in the context.

What is known about reading mode

Content rendering
Content is rendered in a way that is inconsistent and unpredictable. The user cannot count on all-, nor important textual content being made available to them.

Navigation capabilities
The user will not be able to navigate efficiently while using reading mode. They will have to count on having to exit/enter reading mode multiple times while using it to read on different pages.

Tool availability
Users will have to access the reading mode through the various menus in the mobile browser.

Useful features
Users may not have access to features like read aloud and text highlighting when using reading mode. Read aloud is only available on safari and the audio cannot be replayed. There are different fonts and backgrounds to choose from.

What is needed to effectively read [1, 2, 5, 6]

Clear contrasts
There must be a sufficient contrast between text and background for users to be able to discern textual content on the screen.

Plain language
Language use on the interface is simple. Words and sentences are easy to understand [4].

Adjust settings
The user should be able to freely adjust settings to one that suits them the best.

Fonts and layout
Both font size, style and type impact reading and users. As does page layout like text justification, line- height and length and letter spacing. Generally, fonts should be in a basic variant of sans-serif or serif and the layout or interface should not be crowded [6].

Context component

Digital environment
A digital environment such as this website follows many rules but is not entirely dissimilar to physical environments in the sense that there can still be barriers to reaching goals.

A website

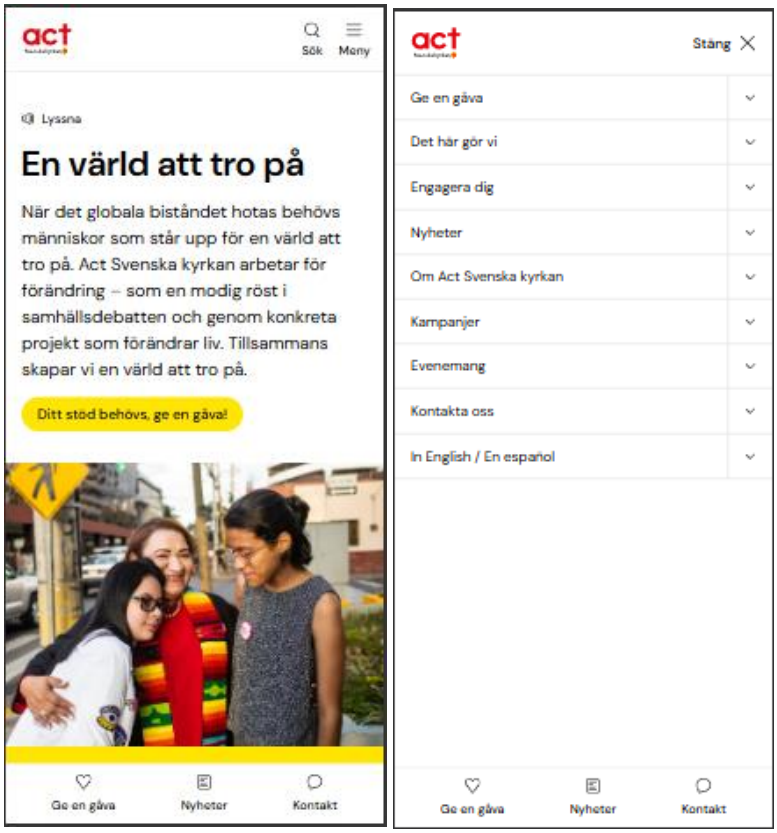
Visuals
Accessibility applies to visuals too. On a website, visuals that impact reading may be similar to readability. Such as colors used on the website, both background and text.

Content
The textual content that is of interest for reading may be articles, blog posts or informative pages. The pages might have a varying amount of content. Images and visualisations provide an alternative means of understanding content.

Accessibility
The content of the website should generally be accessible. The WCAG [8] is used as a means of making a website accessible. A website must be marked up according to best practices to be accessible to everyone. Heading tags should for example follow a strict hierarchy to uphold context of information. Images should have alternative texts.

Act Church of Sweden [9]

This website is the particular context of this study. The reading mode should help users read on this website. Act wants their content to be accessible, and in particular content regarding making a donation (ge en gåva), about what the organization does (om Act svenska kyrkan och Det här gör vi) and how to become an engaged member in the community (engagera dig), for example an Ageravolontär. The long term goal is for all content to be accessible to everyone.



Left: starting page. Right: menu tabs on a mobile browser

The core pages
They are outlined here by content as shown in the menu tabs as well as relevance to the study through the tasks in the heuristic evaluation.

- **Ge en gåva**
    - users can use a form to give a one time or monthly donation as
        - an individual
        - a parish
        - a company
    - additional information include
        - How your gift is handled, how the money that gets donated is allocated, to the charity goal, to administration and control as well as materials and activities
        - What steps Act takes to make sure donations are handled safely to avoid going into the wrong hands
        - Tax reduction on donations, how much you have to donate to be eligible for tax reduction
- **The form for donations appear throughout most pages on the website**
- **Om Act Svenska Kyrkan**
    - What are the parts of the organisation
    - How is it managed

    - Which global partners are part of the collaboration
    - History
- **Det här gör vi**
    - What does Act Church of Sweden do and in which areas to they provide support, each with their own news, examples of supportive actions and stories
        - Catastrophes
        - Provision and climate justice
        - Equality and health
        - Peace and democracy
        - Faith and education
    - Blog posts reporting on charity support actions from different parts of the world
- **Engagera dig**
    - Different ways of becoming engaged in the community
        - Upcoming events
        - Charity campaigns
        - Ageravolontär
            - What it means to volunteer as a Ageravolontär within the organisation or a local parish
            - Why and how to volunteer
            - Blog posts reporting on volunteer work
        - List of example in-person or digital activities to meet and to gather donations
        - Exchange programme

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Appendix K: Overview of comparative analysis



Nielsens 10	Deque systems	conceptual overlap
		Low
		None
		Moderate
		High
		Moderate
		High
		None
		High
		None
		None
		None
		None